

A Modular Bridge From Algebraic to Analytic: Iwasawa Theory à la Ribet

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Abstract. In this article we give Wiles’ proof of the Iwasawa Main Conjecture for totally real fields, specialized to the case over $\mathbb{Q}$. We also develop the necessary machinery not just for the proof, but also to even state the theorem, including brief treatments of p -adic L -functions and Selmer groups, as well as the construction of the Galois representation attached to a Hida family.

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1 Introduction

Many problems in number theory can be described as some variation on the theme of “algebraic equals analytic.” This is somewhat remarkable; when defining structures of arithmetic significance, there is often little reason *a priori* to expect that an object constructed from the algebraic data of one’s situation should carry the same information as something which comes about by piecing together data analytically. Thus, when two such constructions happen to agree, one tends to find a deep theory underlying their equivalence.

In this article, we will prove one such equivalence relating two objects that one can attach to a Dirichlet character ψ : a certain module, defined by interpolating together a large family of ideal class groups related to ψ , and the p -adic Dirichlet L -function for ψ , defined in terms of the L -functions attached to ψ and its various twists. Such a relationship was first conjectured to hold following the work of Iwasawa in the 1960s on p -adic L -functions and the class groups of cyclotomic number fields.

The equivalence of the data contained in these two kinds of objects is the subject of Iwasawa theory, which extends far beyond the case of Dirichlet characters. However, the equivalence in this (relatively) simple case, which is known as the Main Conjecture of Iwasawa theory, was the prototype for all further developments in Iwasawa theory, and the techniques underlying its proof continue to be relevant to date.

We will give a variant of Wiles' 1990 proof presented in [Sk09], treating only the case over $\mathbb{Q}$ rather than the general totally real situation that the proof addresses. Wiles' proof was modeled off of Ribet's proof of the converse to Herbrand's theorem, a classical result relating elements of certain ideal class groups to the mod p vanishing of certain L -values. While the Herbrand-Ribet theorem is not used in the proof of the Main Conjecture, its proof encapsulates the key strategy used by Wiles, so we present it here before delving into Iwasawa theory.

Ribet's strategy on its own is not powerful enough to touch the Main Conjecture; the key ingredient needed to upgrade Ribet's idea is Hida theory, a method for placing modular forms into large families of mod p congruences with other modular forms. This technology, along with the existence of Galois representations attached to modular forms (already used in Ribet's proof) will be enough to finish off the Main Conjecture.

2 Ribet's Converse

Our goal for this section is to prove Ribet's converse to Herbrand's theorem, which gives a relationship between the divisibility of certain Bernoulli numbers and the structure of class groups of cyclotomic number fields. A classical theorem of Clausen and von-Staudt asserts that the denominator of the k th Bernoulli number B_k is the product of all primes p such that $p - 1 \mid k$. Thus, fixing some odd prime p , we have $B_k \in \mathbb{Z}_p$ for even k between 2 and $p - 3$, and we may ask about the p -divisibility of these B_k .

Kummer's criterion for regularity asserts that p divides the class number of $\mathbb{Q}(\mu_p)$ if and only if p divides one of $B_2, B_4, \dots, B_{p-3}$. The combined theorems of Herbrand and Ribet provide a strengthening of this result, showing that the p -divisibility of a particular B_k is equivalent to a particular component of the p -part of the class group of $\mathbb{Q}(\mu_p)$ being nontrivial.

We will prove Ribet's direction of the above equivalence. Let $\chi : G_{\mathbb{Q}} \rightarrow \mathbb{F}_p^\times$ denote the mod- p cyclotomic character of $G_{\mathbb{Q}} = \text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$. Let C be the class group of $\mathbb{Q}(\mu_p)$, a finite abelian group; let C_p be its p -torsion. Then C_p is an $\mathbb{F}_p$ -module with an action of $\Delta := \text{Gal}(\mathbb{Q}(\mu_p)/\mathbb{Q})$, so since this group has order prime to p we can form an isotypic decomposition of C_p in terms of the characters of Δ . Note that Δ is cyclic and has $\mathbb{F}_p$ -character group generated by χ . Then Ribet's converse is

Theorem 1. *Let p be an odd prime, and $3 \leq m \leq p - 2$ be an odd integer. Then $p \mid B_{p-m}$ if and only if $C_p^{\chi^m} \neq 0$.*

2.1 The Strategy

The fundamental first step is to recall that by class field theory, we have an isomorphism

$$\text{Art} : C \rightarrow \text{Gal}(H/\mathbb{Q}(\mu_p))$$

where H is the maximal unramified abelian extension of $\mathbb{Q}(\mu_p)$. Art is equivariant for the action of Δ on both sides, where Δ acts on ideal classes on the left side and acts by conjugation on the right side. Thus, the nonvanishing of $C_p^{\chi^m}$ is equivalent to there being some nontrivial p -subextension of $H/\mathbb{Q}(\mu_p)$ with the correct action of Δ .

To summarize, our goal is now to, provided that $p \mid B_{p-m}$, construct an extension $E/\mathbb{Q}(\mu_p)$ such that

1. $E/\mathbb{Q}(\mu_p)$ is abelian
2. $E/\mathbb{Q}(\mu_p)$ is unramified
3. $p \mid \deg(E/\mathbb{Q}(\mu_p))$
4. Δ acts on $\text{Gal}(E/\mathbb{Q}(\mu_p))$ by χ^m

How can we approach this? We may shift the question to the Galois side; the existence of such an extension is equivalent to the existence of a subquotient of $G_{\mathbb{Q}}$ satisfying certain properties. Inspired by the power of representation theory, it's not unreasonable to guess that we could try to find a Galois representation ρ that cuts out this subquotient in a suitable way. This will be our approximate strategy.

Translating condition 4 to a condition on ρ tells us that we want χ^m to factor through ρ . Ribet's key idea was to try to find such a ρ using the Galois representations attached to modular forms.

2.2 Constructing a Good Eigenform

Recall that the Eisenstein series E_k has ℓ th Fourier coefficient $\ell^{k-1} + 1$ for a prime ℓ . From this, one can show that there is a mod- p Galois representation naturally associated to E_k given by $\chi^{k-1} \oplus 1$. Unfortunately, this reducible representation is not very interesting. Instead, we will do something trickier: we will leverage that $p \mid B_k$ to find an eigenform congruent to E_k modulo p , and then work on the much more interesting representation attached to the eigenform.

The weight- k Eisenstein series of level 1 is given by

$$E_k(z) = \frac{\zeta(1-k)}{2} + \sum_{n=1}^{\infty} \left(e^{2\pi i n z} \sum_{d|n} d^{k-1} \right).$$

For technical reasons (E_2 is not very friendly to work with), we choose some $k \geq 4$ with $k \equiv p-m \pmod{p-1}$, so that $\chi^{1-k} = \chi^m$. From Kummer's congruences, we have nonetheless that $p \mid B_{p-m} \Rightarrow p \mid B_k$; then, since we have $\zeta(1-k) = -\frac{B_k}{k}$, we have $p \mid \frac{\zeta(1-k)}{k}$ in $\mathbb{Z}_p$.

One can check that there is a suitable product G_k of powers of E_4 and E_6 that is of weight k with

$$G_k(z) = 1 + \sum_{n=1}^{\infty} a_n e^{2\pi i n z}$$

where $a_n \in \mathbb{Z}$ for all n . Then, putting

$$F(z) = E_k(z) - \frac{\zeta(1-k)}{2} G_k(z) = \sum_{n=1}^{\infty} b_n e^{2\pi i n z},$$

we have $b_n \in \mathbb{Z}_p$, and since $p \mid \zeta(1-k)$, we have

$$b_n \equiv \sum_{d|n} d^{k-1} \pmod{p}$$

for all n . Note also that F has zero constant term and thus is a cusp form (but F itself is nonzero since, say, b_1 is nonzero mod p).

Now we want to upgrade F to an eigenform so that we may obtain a good Galois representation. To do this, let $\mathbb{T}$ be the $\mathbb{Z}_p$ -algebra generated by the Hecke operators T_ℓ for primes ℓ , where $\mathbb{T}$ acts on $S_k(1, \mathbb{Z}_p)$. One can prove that $\mathbb{T}$ is dual to $S_k(1, \mathbb{Z}_p)$, in the sense that homomorphisms $\phi : \mathbb{T} \rightarrow \overline{\mathbb{Q}}_p$ correspond to normalized eigenforms f_ϕ with $T_\ell f_\phi = \phi(T_\ell) f_\phi$.

Since F is congruent mod p to E_k which is an eigenvector for the T_ℓ , the map $\mathbb{T} \rightarrow \mathbb{F}_p$ given by $T_\ell \mapsto b_\ell$ is a $\mathbb{Z}_p$ -algebra homomorphism, with kernel $\mathfrak{m}$ a maximal ideal. Since $\mathbb{T}$ is reduced, $\mathfrak{m}$ contains a minimal prime $\mathfrak{p}$ that is the kernel of a map $\mathbb{T} \rightarrow \overline{\mathbb{Q}}_p$. Let

$$f = \sum_{n=1}^{\infty} c_n e^{2\pi i n z}$$

be the corresponding eigenform. The Fourier coefficients c_n of f lie in a finite extension L of $\mathbb{Q}_p$; let π be a prime of L lying above p . Then since $\mathfrak{p} \subset \mathfrak{m}$, we have by construction

$$c_\ell \equiv b_\ell \equiv \ell^{k-1} + 1 \pmod{\pi}$$

for all primes ℓ . This will be the eigenform whose attached Galois representation we will study.

2.3 Analyzing the representation and finding an eigenvector

Let $\rho : G_{\mathbb{Q}} \rightarrow \mathrm{GL}(V)$ be the Galois representation attached to the eigenform f , where V is a 2-dimensional vector space over L . We have that ρ is (absolutely) irreducible, and that $\mathrm{Tr}(\rho(\mathrm{Frob}_\ell)) \in \mathcal{O}_L$ with

$$\mathrm{Tr}(\rho(\mathrm{Frob}_\ell)) \equiv \ell^{k-1} + 1 \pmod{\pi}$$

for all primes $\ell \neq p$. (See Theorem 10 for the theorem of Deligne of which the existence of this representation is a consequence.)

We will study the residual representations of ρ associated to various Galois-stable lattices in V ; it is one of these mod- p representations that will give us the extension we want.

Let $\kappa = \mathcal{O}_L/\pi$. For any Galois stable lattice $\mathcal{L} \subset V$, the κ -representation $\bar{\rho}$ of $G_{\mathbb{Q}}$ on $\mathcal{L}/\pi\mathcal{L}$ has

$$\mathrm{Tr}(\bar{\rho}(\mathrm{Frob}_\ell)) = \ell^{k-1} + 1 = \chi^{k-1}(\mathrm{Frob}_\ell) + 1(\mathrm{Frob}_\ell)$$

for all primes $\ell \neq p$. Thus, by the Brauer-Nesbitt Theorem we have

$$\bar{\rho}^{\mathrm{ss}} \cong \chi^{k-1} \oplus 1.$$

Our goal will be to construct a particular lattice so that $\bar{\rho}$ gives a non-split extension of χ^{k-1} and 1; then χ^{k-1} will factor through $\bar{\rho}$ as we want, and we will choose our lattice so that the other properties we desire hold as well.

In Ribet's original proof, there were serious technical difficulties at the prime p : at the time, the theory of representations attached to eigenforms was not sufficient to address these. Fortunately, we have the benefit of having access to future technology, and so can give a more streamlined argument. We will use a theorem of Wiles to first construct a preliminary lattice with the goal of extracting a certain eigenvector. We'll use this eigenvector to later construct a second lattice, in a way that will allow us to bypass problems at p .

The result we will use is the following (cf. Theorem 11):

Theorem 2 (Wiles). *Suppose that f is ordinary at p . Then we have*

$$\rho|_{D_p} \cong \begin{pmatrix} \varepsilon_1 & * \\ & \varepsilon_2 \end{pmatrix}$$

where D_p is a decomposition subgroup for p , and ε_2 is unramified.

Note that since $c_p \equiv p^{k-1} + 1 \equiv 1 \pmod{\pi}$, f is indeed ordinary at p . Now let $\{v_1, v_2\}$ be a basis of V in which $\rho|_{D_p}$ is of the form described in Theorem 2. After replacing v_1 with v_1^{-r} for some r if necessary, we have that v_1, v_2 span an $\mathcal{O}_L$ -lattice $\mathcal{L}$ that is stable for $\rho(D_p)$. For convenience, just for the rest of this section let $\bar{\rho}$ denote the representation of D_p on $\mathcal{L}/\pi\mathcal{L}$.

We know that

$$\mathrm{Tr}(\bar{\rho}) = \varepsilon_1 + \varepsilon_2 \equiv 1 + \chi^{k-1} \pmod{\pi}.$$

Now, take some $\sigma_0 \in I_p$ (where I_p is the inertia group for p) such that $\chi^{k-1}(\sigma_0) \neq 1$, which is possible since χ is totally ramified and $p-1 \nmid k-1$. Since ε_2 is unramified, we have

$$\varepsilon_1(\sigma_0) + 1 \equiv \chi^{k-1}(\sigma_0) + 1 \pmod{\pi} \implies \varepsilon_1(\sigma_0) \not\equiv 1 \pmod{\pi}.$$

Thus, we see that v_1 is an eigenvector for $\rho(\sigma_0)$ whose corresponding eigenvalue is not congruent to 1 modulo π . In fact, we see that we must have $\varepsilon_1|_{I_p} \equiv \chi^{k-1} \pmod{\pi}$.

2.4 The lattice and the extension

We are now ready to construct the representation we want. With $v \in V$ an eigenvector for $\rho(\sigma_0)$ as constructed above, define a lattice

$$M = \mathcal{O}_L[G_{\mathbb{Q}}]v \subset V.$$

Note that this is indeed a full rank lattice: its L -span is 2-dimensional since ρ is irreducible, and it is a lattice since $G_{\mathbb{Q}} \times \mathcal{O}_L v$ is compact. From now on let $\bar{\cdot}$ denote reduction to $M/\pi M$.

Now we have the residual representation $\bar{M} = M/\pi M$, which, since $\bar{\rho}^{ss} \simeq \chi^{k-1} \oplus 1$, must be of one of the following forms:

$$\begin{pmatrix} \chi^{k-1} & * \\ & 1 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 1 & * \\ & \chi^{k-1} \end{pmatrix}.$$

We will show that the latter is correct. Write

$$\bar{\rho} = \begin{pmatrix} \theta_1 & * \\ & \theta_2 \end{pmatrix},$$

so either $\theta_1 = \chi^{k-1}$ and $\theta_2 = 1$, or $\theta_1 = 1$ and $\theta_2 = \chi^{k-1}$. In either case, we know $\bar{v}$ cannot be the θ_1 -eigenvector, since v generates $\bar{M}$ over $\mathcal{O}_L[G_{\mathbb{Q}}]$. In particular, $*$ is nontrivial, and $\bar{M}$ is not split. Thus, letting u be the simultaneous eigenvector, $\{u, v\}$ is a basis for $\bar{M}$ making $\bar{\rho}$ into the above form (where we may need to change $*$, but in any case it remains nontrivial).

Now observe that

$$\begin{aligned} \bar{\rho}(\sigma_0)\bar{v} &= \chi^{k-1}(\sigma_0)\bar{v} \neq 1(\sigma_0)\bar{v} \\ &= \theta_2(\sigma_0)\bar{v} + *u. \end{aligned}$$

Thus, we must have $\theta_2 = \chi^{k-1}$ and $\theta_1 = 1$, as we wanted.

Let c be the top left matrix coefficient for $\bar{\rho}$, a function $G_{\mathbb{Q}} \rightarrow \kappa$ (i.e. we have now decided that $*$ is worthy of a permanent name). Observe that we have

$$\begin{pmatrix} 1 & c(\sigma) \\ \chi^{k-1}(\sigma) & * \end{pmatrix} \begin{pmatrix} 1 & c(\tau) \\ \chi^{k-1}(\tau) & * \end{pmatrix} = \begin{pmatrix} *c(\tau) + \chi^{k-1}(\tau)c(\sigma) & * \\ & * \end{pmatrix} \Rightarrow c(\tau) + \chi^{k-1}(\tau)c(\sigma) = c(\sigma\tau).$$

Thus, letting $c'(\sigma) = \chi^{1-k}(\sigma)c(\sigma)$, we see that c' is a cocycle in $Z^1(G_{\mathbb{Q}}, \kappa(\chi^{1-k}))$. c' is homologically nontrivial since $\bar{M}$ is non-split. Moreover, c' is unramified at primes $\ell \neq p$ since ρ is, and so in particular $c(\sigma) = 0$ for $\sigma \in I_{\ell}$. c' is also unramified at p , since by Theorem 2 and our work on the previous lattice, $\rho(I_p)$ has a 1-dimensional invariant subspace with eigenvalues congruent to χ^{k-1} modulo π , so $\bar{M}$ must be split for I_p since $\theta_1 \neq \chi^{k-1}$.

Thus, we have constructed a nontrivial element of Galois cohomology $c' \in H^1(\mathbb{Q}, \kappa(\chi^{1-k}))$. Equivalently, since $G_{\mathbb{Q}(\mu_p)}$ acts trivially on $\kappa(\chi^{1-k})$, this gives a nontrivial homomorphism of groups $G_{\mathbb{Q}(\mu_p)} \rightarrow \kappa(\chi^{1-k})$ that is equivariant for the action of $\text{Gal}(\mathbb{Q}(\mu_p)/\mathbb{Q})$ on both sides. But now we have a quotient of $G_{\mathbb{Q}(\mu_p)}$ which is abelian, p -torsion, kills inertia, and has $\text{Gal}(\mathbb{Q}(\mu_p)/\mathbb{Q})$ -action given by $\chi^{1-k} = \chi^m$. Thus, we've finally obtained an extension of $\mathbb{Q}(\mu_p)$ with all the properties we wanted. This concludes the proof of Ribet's converse.

3 p -adic L -functions

In this section we summarize the basic results in the theory of p -adic L -functions. To do this, we will introduce the notion of p -adic measures, which in some sense is the correct way to formulate a family of L -values; nevertheless, we will then show how the existence of a certain measure implies the existence of analytic p -adic L -functions.

3.1 p -adic measures

Let R be a subring of $\bar{\mathbb{Q}}_p$. For a profinite abelian group G , we write $\mathcal{C}(G, R)$ for the space of all continuous functions $G \rightarrow R$, topologized with the sup norm coming from the p -adic norm on $\bar{\mathbb{Q}}_p$. We then define $\mathcal{M}(G, R)$ to be the space $\text{hom}_{cts}(\mathcal{C}(G, R), R)$ of bounded linear functionals on $\mathcal{C}(G, R)$. We refer to elements of $\mathcal{M}(G, R)$ as p -adic measures, or simply measures, on G . For $\mu \in \mathcal{M}(G, R)$ and $f \in \mathcal{C}(G, R)$, we typically write

$$\int_G f d\mu$$

for the value of μ on f . We will usually take $G = \mathbb{Z}_p$ or $\mathbb{Z}_p^\times$ or a subgroup thereof. In the literature, it is also common to take G to be a pro- p Galois group; it is enlightening to consider how this perspective connects sections 3 and 4.

A fundamental result in the analysis of p -adic functions is the following:

Theorem 3 (Mahler). *Let R be a subring of $\bar{\mathbb{Q}}_p$, and let $f : \mathbb{Z}_p \rightarrow R$ be any function. Then f is continuous if and only if there is a sequence $\{a_n\}_{n=0}^\infty \subset R$ with $a_n \rightarrow 0$ so that*

$$f(x) = \sum_{n=0}^{\infty} a_n \binom{x}{n}$$

where the convergence is uniform in x . Also, in this case the sequence a_n is unique.

Here by $\binom{x}{n}$ we mean the usual binomial coefficient $\frac{x(x-1)\dots(x-n+1)}{n!}$. The proof is elementary and can be found in [Hi93, §3.2].

From Mahler's Theorem, we see that each element $\mu \in \mathcal{M}(\mathbb{Z}_p, \mathbb{Z}_p)$ is determined by the values $\int \binom{x}{n} d\mu$ for $n \geq 0$. In this way we may associate to μ a unique element of $\mathbb{Z}_p[[T]]$ via

$$\mu \mapsto \sum_{n=0}^{\infty} \left(\int_{\mathbb{Z}_p} \binom{x}{n} d\mu(x) \right) T^n.$$

On the other hand, for any $\sum_{n=0}^{\infty} b_n T^n \in \mathbb{Z}_p[[T]]$, we may obtain a measure μ on $\mathbb{Z}_p$ by prescribing

$$\int_{\mathbb{Z}_p} \binom{x}{n} d\mu(x) = b_n$$

which is automatically bounded since each $b_n \in \mathbb{Z}_p$. Thus we obtain a bijection $\mathcal{M}(\mathbb{Z}_p, \mathbb{Z}_p) \xrightarrow{\sim} \mathbb{Z}_p[[T]]$. Moreover, from this it is easy to see that we obtain a bijection $\mathcal{M}(\mathbb{Z}_p, \mathcal{O}) \xrightarrow{\sim} \mathcal{O}[[T]]$ for $\mathcal{O}$ the ring of integers of a finite extension of $\mathbb{Q}_p$. It is clear that this map is $\mathcal{O}$ -linear, and in fact more is true; for measures μ, ν , it can be checked with relative ease that the convolution product $\mu * \nu$ given by

$$\int_{\mathbb{Z}_p} f d(\mu * \nu) = \iint_{\mathbb{Z}_p \times \mathbb{Z}_p} f(x+y) d\mu(x) d\nu(y)$$

makes this map into a ring isomorphism. More generally, we have

Proposition 1. *For an abelian profinite group G there is a ring isomorphism*

$$\mathcal{M}(G, \mathcal{O}_L) \simeq \varprojlim_H \mathcal{O}_L[G/H]$$

where H ranges over open subgroups of G , and the ring structure on the left is given by convolution.

It is of great relevance to the broader theory that this identifies measures on $\mathbb{Z}_p$ with the Iwasawa algebras $\Lambda_{\mathcal{O}}$ to be introduced in Section 4. For a proof, see [Co06, §3.2].

For technical reasons (to be justified shortly), to a measure μ we will actually attach the power series

$$\Phi_{\mu}(T) = \sum_{n=0}^{\infty} \left(\int_{\mathbb{Z}_p} \binom{x}{n} d\mu(x) \right) (T-1)^n$$

expanded about $1 \in \mathcal{O}$. Since Φ_{μ} has coefficients in $\mathcal{O}$, we see that $\Phi_{\mu}(T)$ is convergent on the region $|T-1| < 1$; e.g., for $\mathcal{O} = \mathbb{Z}_p$, Φ_{μ} converges on $1 + p\mathbb{Z}_p$.

Recall that for $x \in \mathbb{Z}_p$ and $a \in 1 + p\mathbb{Z}_p$, we may define the exponential $a^x \in 1 + p\mathbb{Z}_p$ by

$$a^x = \sum_{n=0}^{\infty} \binom{x}{n} (a-1)^n.$$

Then by our definition of Φ_{μ} , we have the nice identity

$$\int_{\mathbb{Z}_p} a^x d\mu(x) = \int_{\mathbb{Z}_p} \sum_{n=0}^{\infty} \binom{x}{n} (a-1)^n d\mu(x) = \Phi_{\mu}(a)$$

where we use the boundedness of μ to exchange the sum and the integral. Slightly more generally, we record

Lemma 1. *If $f(x) = a^x$ for $a \in 1 + p\mathbb{Z}_p$, then $\Phi_{f\mu}(T) = \Phi_{\mu}(aT)$. In particular, this applies when ψ is a Dirichlet character of p -power conductor and p -power order.*

Proof. If $f(x) = a^x$, then we have for $b \in 1 + p\mathbb{Z}_p$

$$\Phi_{f\mu}(b) = \int_{\mathbb{Z}_p} a^x d(f\mu)(x) = \int_{\mathbb{Z}_p} (ab)^x d\mu(x) = \Phi_{\mu}(ab),$$

from which we see $\Phi_{f\mu}(T) = \Phi_{\mu}(aT)$. When ψ is a Dirichlet character of the above form, there is some p -power root of unity ζ so that $\psi(x) = \zeta^x$ for all $x \in \mathbb{Z}_p$, from which the claim follows. $\square$

3.2 Pseudo-measures attached to Dirichlet L -functions

Our goal is to define a p -adic analytic object which interpolates the values of Dirichlet L -functions $L(s, \psi)$ at negative integers. Rather than giving a literal analytic function on $\mathbb{Z}_p$ (or extensions thereof) that achieve this, we will first give a manifestation of such an object as a p -adic measure, which is in many ways more natural.

For our purposes, we will need to slightly expand the class of measures we consider to include what we will call pseudo-measures, which were first introduced by Serre. This is necessary essentially because of the pole of the L -function of the trivial character (i.e. the Riemann zeta function) at $s = 1$; when considering a Dirichlet character with non-trivial tame character (i.e. whose prime to p part is nontrivial), one may ignore all the forthcoming discussion of pseudo-measures and instead work only with the genuine measures previously discussed.

We use the notation of [Co06]. Consider the fraction field $F = \text{Frac}(\mathcal{M}(G, \mathcal{O}_L))$. We define an $\mathcal{O}_L$ -valued pseudo-measure on G to be an element $\mu \in \text{Frac}(\mathcal{M}(G, \mathcal{O}_L))$ such that

$$(g-1)\mu \in \mathcal{M}(G, \mathcal{O}_L)$$

for all $g \in G$. There is not a natural notion of integrating a general function against a pseudo-measure; however, note that a homomorphism $\phi : G \rightarrow \bar{\mathbb{Q}}_p^{\times}$ may be extended to a homomorphism $\mathcal{M}(G, \mathcal{O}_L) \rightarrow \bar{\mathbb{Q}}_p^{\times}$ by putting

$$\phi(\mu) = \int_G \phi d\mu.$$

Then for a pseudo-measure μ , if ϕ is non-trivial, then we may define

$$\int_G \phi d\mu = \frac{\int_G \phi d((g-1)\mu)}{\phi(g) - 1}$$

for some $g \in G$ for which $\phi(g) \neq 1$; this is independent of g since, as seen above, ϕ is a homomorphism out of $\mathcal{M}(G, \mathcal{O}_L)$. This will be sufficient for our purposes.

The following remarkable result traces back to Kubota and Leopoldt, who were the first to construct an interpolation of L -values as a single p -adic L -function. The reformulation in terms of p -adic valuations was given by Mazur, and is a significant strengthening:

Theorem 4. *Let $\psi = \theta\eta$ be a primitive Dirichlet character where θ is primitive of conductor N prime to p and η is primitive of conductor p^n (where n may be zero). Then there is a unique $\mathbb{Z}_p[\psi]$ -valued pseudo-measure ζ_θ satisfying*

$$\int_{\mathbb{Z}_p^\times} \eta(x) x^k d\zeta_\theta(x) = (1 - \psi(p)p^{k-1})L(1 - k, \psi)$$

for all integers $k > 0$. Moreover, if θ is nontrivial then ζ_θ is a genuine measure.

In particular, for any character θ of conductor prime to p , the (pseudo-)measure ζ_θ encodes *all* the non-positive integer L -values for *all* twists of ψ by p -power order characters. For a proof of this great theorem, see [La90].

3.3 From measures to analytic functions

Equipped with the p -adic measures ζ_θ , we now show how from these we can obtain genuine p -adic analytic functions that interpolate L -values. These will be the data that appear on the analytic side of the Main Conjecture. For a Dirichlet character ψ as in Theorem 4, we would like to replace the integer k with a p -adic variable, making the definition

$$L_p(s, \psi) \stackrel{?}{=} \int_{\mathbb{Z}_p^\times} \eta(x) x^{1-s} d\zeta_\theta(x).$$

Unfortunately, this definition does not make sense - for general $s \in \mathbb{Z}_p$ and $x \in \mathbb{Z}_p^\times$, the exponential x^s need not be defined. This exponential *is* well defined when $x \in 1 + p\mathbb{Z}_p$; thus, taking a suitable normalization of the above fake definition will give us a well-defined notion. We define the character

$$\langle \cdot \rangle : \mathbb{Z}_p^\times \rightarrow 1 + p\mathbb{Z}_p, \quad \langle x \rangle := \omega(x)^{-1}x.$$

Then, we define the p -adic Dirichlet L -function for ψ to be

$$L_p(s, \psi) := \int_{\mathbb{Z}_p^\times} \eta(x) \langle x \rangle^{1-s} d\zeta_\theta(x).$$

Note then that we have the interpolation formulae

$$L_p(1 - k, \psi) = (1 - \psi\omega^{-k}(p)p^{1-k})L(1 - k, \psi\omega^{-k})$$

for all $k > 0$. The apparent defect of needing the ω -power twists is a consequence of the failure of the exponential to be defined on all of $\mathbb{Z}_p^\times$, which in turn can be seen as a consequence of $\mathbb{Z}_p^\times$ consisting of $p-1$ copies of the p -adic unit ball. In this sense, to define p -adic L -functions interpolating the values of $L(s, \psi)$, we in fact need to define $p-1$ different “branches” corresponding to these $p-1$ balls.

Also, note that for ψ an odd Dirichlet character (i.e. $\psi(-1) = -1$), we have $L_p(1-k, \psi) = 0$ for all $k > 0$, so $L_p(s, \psi)$ is identically zero in this case. For this reason, for the remainder of this section we specialize to the case of ψ even (though our statements remain vacuously true in the other case).

Now we will show how these $L_p(s, \psi)$ can be viewed as analytic power series with certain evaluation properties. Define a (pseudo-)measure μ_ψ on $1 + p\mathbb{Z}_p$ by

$$\int_{1+p\mathbb{Z}_p} f(x) d\mu_\psi := \int_{\mathbb{Z}_p^\times} \eta(x) \omega^{-1}(x) x f(\langle x \rangle) d\zeta_\theta(x)$$

which is a measure if θ is non-trivial. Let $\iota : \mathbb{Z}_p \xrightarrow{\sim} 1 + p\mathbb{Z}_p$ be the isomorphism given by $z \mapsto (1+p)^z$. Then we have

$$\begin{aligned} L_p(s, \psi) &= \int_{\mathbb{Z}_p^\times} \eta(x) \omega^{-1}(x) x \langle x \rangle^{-s} d\zeta_\theta(x) \\ &= \int_{1+p\mathbb{Z}_p} y^{-s} d\mu_\psi(y) \\ &= \int_{\mathbb{Z}_p} (1+p)^{-zs} d(\iota^* \mu_\psi)(z) \\ &= \Phi_{\iota^* \mu_\psi}((1+p)^{-s}), \end{aligned}$$

where the last equality follows from Lemma 1. Note that if μ_ψ is a pseudo-measure then by $\Phi_{\iota^* \mu_\psi}$ we really mean an appropriate quotient of power series. Thus, we have realized $L_p(s, \psi)$ as the composition of the function $s \mapsto (1+p)^{-s}$, analytic on all of $\mathbb{Z}_p$, with the (quotient of) power series $\Phi_{\iota^* \mu_\psi}$ about $1 \in 1 + p\mathbb{Z}_p$ which is meromorphic with at worst a simple pole at $1+p$. Thus we see that $L_p(s, \psi)$ is analytic on $\mathbb{Z}_p$ if θ is nontrivial, and is meromorphic with simple pole at 1 otherwise.

However, this meromorphic function remembers more than simply its evaluation at points of $\mathbb{Z}_p$; i.e., $L_p(s, \psi)$ shouldn't just see the L -values $L(1-k, \psi)$, and indeed should remember slightly more L -values coming from certain p -power twists of ψ , corresponding to how the measure from which $L_p(s, \psi)$ was derived saw a much larger family of L -values than just those for ψ .

We make this precise as follows. Let ξ be a Dirichlet character of p -power conductor and p -power order. In particular, ξ factors through $1 + p\mathbb{Z}_p$. Since ι is an isomorphism $\mathbb{Z}_p \rightarrow 1 + p\mathbb{Z}_p$, we may freely identify $L(s, \psi)$ with $\Phi_{\iota^* \mu_\psi}$, which corresponds uniquely to $\iota^* \mu_\psi$, which in turn corresponds uniquely to μ_ψ (again since ι is an isomorphism). Thus, there is a clear notion of “evaluating $L_p(s, \psi)$ at characters of $1 + p\mathbb{Z}_p$ ” other than $y \mapsto y^{-s}$; in particular, we may evaluate $L_p(s, \psi)$ at $y \mapsto \xi(y)y^{-s}$. Since ξ factors through $1 + p\mathbb{Z}_p$, we in fact have

$$\int_{1+p\mathbb{Z}_p} \xi(y) y^{-s} d\mu_\psi = \int_{\mathbb{Z}_p^\times} \eta \xi(x) \langle x \rangle^{1-s} d\zeta_\theta(x) = L_p(s, \psi \xi).$$

In this way, by viewing $L(s, \psi)$ as its corresponding measure μ_ψ , we see that $L(s, \psi)$ records the values $L_p(s, \psi \xi)$.

This can be phrased more algebraically. Let $\mathcal{O}$ be $\mathbb{Z}_p$ adjoined with the values of ψ . If we view $L_p(s, \psi)$ as living in $\mathcal{M}(\mathbb{Z}_p, \mathcal{O}) \simeq \mathcal{O}[[T-1]]$, then the values of the many characters of $\mathcal{O}[[T-1]]$ on $L_p(s, \psi)$ recover not just the L -values $L(1-k, \psi)$, but also the values of $L(1-k, \psi \xi)$.

This discussion is all kosher if θ is non-trivial; however, if θ is trivial and we are dealing with pseudo-measures, then we must recall that ζ_θ cannot be evaluated at the trivial character. In particular, if $\psi = \eta = \xi^{-1}$, then we do not obtain a well-defined evaluation as above when $s = 1$ (which agrees with the pole of the Riemann zeta function at $s = 1$). Algebraically, this means that the denominator of $L_p(s, \psi)$ when viewed in $\text{Frac}(\mathcal{O}[[T-1]])$ lies in the kernel of the character corresponding to twisting by ξ and evaluating at $s = 1$. Using Lemma 1, we see this character is given by $T \mapsto \zeta(1+p)^{-1}$, so this kernel is generated by

$$T - \zeta(1+p)^{-1} = (T-1) + (1 - \zeta(1+p)^{-1}).$$

Since $L_p(s, 1)$ has simple pole at $s = 1$, this denominator must be a unit multiple of $T - \zeta(1+p)^{-1}$.

The latter algebraic viewpoint shall be the perspective used in the coming sections, and accordingly for the rest of this article we will view p -adic L -functions as “analytic functions in $\mathcal{O}[[T-1]]$ ” rather than as “measures in $\mathcal{M}(\mathbb{Z}_p, \mathcal{O})$ ”. However, we hope that this section (and especially Theorem 4) has made the power and versatility of the measure theoretic approach somewhat clear.

Now we fix some notation to be used for the rest of this article. For ζ a primitive p^t th root of unity, let ψ_ζ denote the mod p^t Dirichlet character which sends $1+p \in \mathbb{Z}_p$ to ζ . For $k \geq 0$ and ζ a p -power root of unity, let $\phi_{k,\zeta}$ be the character of $\mathcal{O}[[T-1]]$ given by $T \mapsto \zeta(1+p)^k$. Note that in our identification between $\mathbb{Z}_p$ and $1+p\mathbb{Z}_p$ we have $T \mapsto (1+p)^{-s}$, and so the k th power on the $(1+p)$ corresponds to taking $s = -k$. We will define $\mathcal{G}_\psi, \mathcal{H}_\psi \in \mathcal{O}[[T-1]]$ by

$$\mathcal{H}_\psi := \begin{cases} T - \zeta(1+p)^{-1} & \psi = \psi_\zeta \\ 1 & \text{otherwise,} \end{cases}$$

and then defining $\mathcal{G}_\psi$ so that

$$\frac{\phi_{k,\zeta}(\mathcal{G}_\psi)}{\phi_{k,\zeta}(\mathcal{H}_\psi)} = L_p(-k, \psi\psi_\zeta) = L^{\{p\}}(-k, \psi\psi_\zeta \omega^{-(k+1)}),$$

where $L^{\{p\}}(s, \psi)$ denotes the usual Dirichlet L -function for ψ with the p -factor removed. That this $\mathcal{G}_\psi$ is unique and well-defined is an immediate consequence of the above discussion.

The following calculation will be needed in our proof of the Main Conjecture:

Lemma 2. *Suppose that ψ is a primitive Dirichlet character with p -power order and conductor. Then $\mathcal{G}_\psi$ is a unit in $\mathcal{O}[[T-1]]$.*

Proof. Let ζ be so that $\psi = \psi_\zeta$. Then by Lemma 1 the evaluation map $\phi_{1,\zeta^{-1}}$ corresponds to performing an invertible linear change of coordinates $T-1$ to $\zeta T-1$, and then taking $\zeta T-1 = 0$. To show $\mathcal{G}_\psi$ is a unit, it then suffices to show that when expanded as a power series in $\zeta T-1$ that $\mathcal{G}_\psi$ has unit constant term; i.e., we must show that $\phi_{1,\zeta^{-1}}(\mathcal{G}_\psi)$ is a p -adic unit. For this, see that we have

$$\frac{\phi_{1,\zeta^{-1}}(\mathcal{G}_\psi)}{\phi_{1,\zeta^{-1}}(\mathcal{H}_\psi)} = L_p(0, \psi\psi_\zeta^{-1}) = L_p(0, \mathbf{1}) = L^{\{p\}}(0, \omega^{-1}) = L(0, \omega^{-1}).$$

We can see from the above that $\mathcal{H}_\psi$ has constant term a uniformizer of $\mathcal{O}$ when expanded in $\zeta T-1$, so it suffices to show that $L(0, \omega^{-1})$ is not a p -adic integer. For this we use the formula

$$L(0, \omega^{-1}) = \frac{-1}{\tau(\omega^{-1})\sqrt{p}} \sum_{a=1}^p \omega^{-1}(a)a,$$

where $\tau(\omega^{-1})$ denotes the Gauss sum for ω^{-1} . Then, since $\omega(a) \equiv a \pmod{p}$, we have

$$\sum_{a=1}^p \omega^{-1}(a)a \equiv p-1 \pmod{p},$$

from which the desired result follows. $\square$

For the rest of this article, we use a slightly different convention for p -adic L -functions. We shall from now on make the change of coordinates $T-1 \mapsto T$, so $\mathcal{G}_\psi$ and $\mathcal{H}_\psi$ will be viewed as elements of $\mathcal{O}[[T]]$ rather than of $\mathcal{O}[[T-1]]$. For the sake of simplifying some formulae later on, we will put

$$g_\psi := \mathcal{G}_{\omega\psi}, \quad h_\psi := \mathcal{H}_{\omega\psi}.$$

Then we have the interpolation formulae

$$\frac{\phi_{k,\zeta}(g_\psi)}{\phi_{k,\zeta}(h_\psi)} = L_p(-k, \omega\psi\psi_\zeta) = L^{\{p\}}(-k, \psi\psi_\zeta\omega^{-k}).$$

In addition to making these evaluations slightly nicer, this convention also arises naturally via different approaches to defining p -adic L -functions (e.g. that of [Wa97]) which we do not discuss here. For us, this basically amounts to absorbing the ω coming from the $\langle x \rangle^{1-s}$ in our p -adic integral definition of $L_p(s, \psi)$ into ψ .

Note that now even Dirichlet characters are the ones whose p -adic L -functions vanish identically, and so we will primarily focus our attention to odd characters. In particular, our statement of the Main Conjecture (and all related discussion) will be for odd Dirichlet characters ψ .

4 Class Groups and Selmer Groups

Now we begin with the algebraic preliminaries needed to state the Iwasawa Main Conjecture. We will start by building toward the original formulation of the Main Conjecture coming from the study of class groups of cyclotomic fields. This is the form in which Mazur and Wiles originally proved the conjecture, as well as the form which Wiles' result for totally real fields addresses. However, we will restate the conjecture in a slightly different form using the modern language of Selmer groups; we will prove that this formulation implies Iwasawa's original conjecture, and then prove this modern statement in Section 6.

4.1 The Iwasawa Algebra Λ

Consider the infinite extension $\mathbb{Q}(\mu_{p^\infty}) := \bigcup_{n=1}^\infty \mathbb{Q}(\mu_{p^n})$ of $\mathbb{Q}$. The p -adic cyclotomic character χ factors through $\text{Gal}(\mathbb{Q}(\mu_{p^\infty})/\mathbb{Q})$, giving an isomorphism of this group with $\mathbb{Z}_p^\times$. Under the decomposition $\mathbb{Z}_p^\times = \mu_{p-1} \times (1 + p\mathbb{Z}_p)$, the quotient of $\text{Gal}(\mathbb{Q}(\mu_{p^\infty})/\mathbb{Q})$ corresponding to μ_{p-1} is $\text{Gal}(\mathbb{Q}(\mu_p)/\mathbb{Q})$. Then let $\mathbb{Q}_\infty$ be the fixed field of μ_{p-1} , so that

$$\mathbb{Z}_p \cong 1 + p\mathbb{Z}_p \cong \text{Gal}(\mathbb{Q}_\infty/\mathbb{Q}) =: \Gamma.$$

Note then that the closed subgroups of Γ correspond to the subgroups $p^n\mathbb{Z}_p$ of $\mathbb{Z}_p$.

$\mathbb{Q}_\infty/\mathbb{Q}$ is the so-called ‘‘cyclotomic $\mathbb{Z}_p$ -extension’’ of $\mathbb{Q}$, and is the prototype for $\mathbb{Z}_p$ -extensions in general (that is, extensions of a number field with Galois group isomorphic to $\mathbb{Z}_p$). Putting $\mathbb{Q}_n := (\mathbb{Q}(\mu_{p^n}))^{\mu_{p-1}}$, the $\mathbb{Q}_n$ correspond to the closed subgroups Γ_n of Γ coming from $p^n\mathbb{Z}_p$.

Put $K = \mathbb{Q}(\mu_N)$. Then we may put $K_\infty = K\mathbb{Q}_\infty$ and $K_n = K\mathbb{Q}_n$ so that K_∞/K is a $\mathbb{Z}_p$ -extension with Galois group $\Gamma_K \subset \Gamma$. Iwasawa sought to compute the p -part of the class groups of the K_n , and more generally the class groups of the fields in the intermediary tower of any $\mathbb{Z}_p$ -extension. He did this by considering a module structure on the Galois group of an extension related to K_∞/K . This method, which we now describe, will be the foundation for our approach to the main conjecture, and in fact is the key idea which underpins all of Iwasawa theory.

Let L_n be the maximal unramified abelian p -extension of K_n , so

$$X_n := \text{Gal}(L_n/K_n) \cong \text{Cl}(K_n)[p^\infty]$$

(p -primary part of the class group). Then writing $L_\infty = \bigcup_{n=1}^\infty L_n$, we have that L_∞ is the maximal unramified abelian pro- p extension of K_∞ , and we may consider the abelian group

$$X_\infty := \text{Gal}(L_\infty/K_\infty).$$

Note that since L_n/K_n is abelian, we have a conjugation action of $\Gamma/\Gamma_n = \text{Gal}(K_n/K)$ on X_n . Thus, since X_∞ is pro- p , we have a continuous action of the completed group ring $\Lambda := \mathbb{Z}_p[[\Gamma]]$ on $\varprojlim X_n = X_\infty$.

It is known that Λ is isomorphic to $\mathbb{Z}_p[[T]]$, which is a complete Noetherian regular local ring of dimension 2. In particular, Λ is a UFD. The same is true of the rings $\Lambda_{\mathcal{O}} := \Lambda \otimes_{\mathbb{Z}_p} \mathcal{O}$ for $\mathcal{O}$ the ring of integers of a finite extension of $\mathbb{Q}_p$. $\Lambda_{\mathcal{O}}$ enjoys a number of nice structural properties which we will use freely. Most notably, we have

Proposition 2 (Weierstrass Preparation Theorem). *Let $f(T) \in \mathcal{O}[[T]]$ be nonzero for $\mathcal{O}$ the ring of integers of a finite extension of $\mathbb{Q}_p$. Fix a uniformizer π of $\mathcal{O}$. Then we have unique $p(T) \in \mathcal{O}[T]$, $u(T) \in \mathcal{O}[[T]]^\times$, and $t \geq 0$ so that*

$$f(T) = \pi^t p(T) u(T).$$

Moreover, p is distinguished; that is, p is a monic polynomial whose non-leading coefficients are divisible by π .

Proposition 3. *Let M be a finitely generated $\Lambda_{\mathcal{O}}$ -module. Then there exist distinguished irreducible $f_1, \dots, f_n \in \Lambda_{\mathcal{O}}$ and $r \geq 0$ so that there is a $\Lambda_{\mathcal{O}}$ -map*

$$M \rightarrow \Lambda_{\mathcal{O}}^r \oplus \bigoplus_{i=1}^n \Lambda_{\mathcal{O}} / (f_i^{n_i})$$

with finite kernel and cokernel.

The above decomposition is unique up to permuting the factors and units. In particular, the ideal of $\Lambda_{\mathcal{O}}$ generated by the product f of the f_i 's is uniquely determined; we call (f) the *characteristic ideal* $Ch(M)$ of M . Note that $Ch(M)$ is nonzero if and only if M is a torsion $\Lambda_{\mathcal{O}}$ -module. This is a very important invariant of Λ -modules that will be discussed further in the next subsection. For a proof of these facts and more generalities on Λ -modules, see [Wa97, §13.2].

Iwasawa proved that X_∞ is a finite Λ -module, and then used the above structure theorem along with the arithmetic of $\mathbb{Z}_p$ -extensions to prove asymptotics (in terms of the f_i appearing in the decomposition) on the size of $X_n = \text{Cl}(K_n)[p^\infty]$. More specifically, one can find certain projectors $\nu_n \in \Lambda$ so that, for n sufficiently large, $X/\nu_n \simeq X_n$; then, using the structure theorem, to obtain an asymptotic it suffices to show that X_∞ is a torsion Λ -module, and then compute $\#(\Lambda/(f_i, p_n))$. To prove this result in full here would take us too far astray from our journey toward the Main Conjecture, but see [Wa97, §13].

The key insight in Iwasawa's proof was that in order to understand each X_n , it was sufficient to interpolate these groups into a larger p -adic object, and then use this additional structure to produce statements about all the X_n simultaneously. This philosophy will, in various forms, underlie all of our subsequent constructions and proof strategies.

We will prove a lemma on Λ -modules that we will need later:

Lemma 3. *Let $\mathcal{O}$ be the ring of integers of a finite extension of $\mathbb{Q}_p$ and let M be a finite $\Lambda_{\mathcal{O}}$ -module containing no finite order submodules. Then for nonzero $r \in \Lambda_{\mathcal{O}}$ we have*

$$\#M/rM = \#\Lambda_{\mathcal{O}}/(r, Ch(M)).$$

Proof. For convenience write Λ for $\Lambda_{\mathcal{O}}$. If M is not torsion then both sides are infinite, so assume M is torsion. By Theorem 3, there is a Λ -map

$$M \rightarrow \prod_{i=1}^n \Lambda / (f_i^{n_i})$$

with finite kernel and cokernel, which we denote by $\ker$ and coker respectively. By assumption on M , $\ker = 0$, so we have an exact sequence

$$0 \rightarrow M \rightarrow \prod \Lambda / (f_i^{n_i}) \rightarrow \text{coker} \rightarrow 0.$$

Multiplying this sequence by r and invoking the snake lemma then gives

$$0 \rightarrow M[r] \rightarrow \prod \Lambda / (f_i^{n_i})[r] \rightarrow \text{coker}[r] \rightarrow M/rM \rightarrow \prod \Lambda / (f_i^{n_i}, r) \rightarrow \text{coker}/r \rightarrow 0.$$

Since the f_i are irreducible, if $\prod \Lambda / (f_i^{n_i})[r]$ is nonzero, then r must be divisible by some of the f_i . Suppose that $r = f_i^a g$ where $(f_i, g) = 1$, and we assume without loss of generality that $n_i > a$. Then we have an exact sequence

$$0 \rightarrow \Lambda/(f_i^{n_i-a}, g) \rightarrow \Lambda/(f_i^{n_i}, f_i^a g) \rightarrow \Lambda/(f_i^a) \rightarrow 0.$$

The first term and the last term are finite and infinite, respectively; see [Wa97][§13.2]. It follows that $\Lambda/(f_i^{n_i}, r)$ is infinite, whence M/rM is also infinite since $\text{coker}[r]$ and coker/r are finite order. Otherwise, $\prod \Lambda/(f_i^{n_i})[r] = 0$. In this case, we have $\#\text{coker}[r] = \#\text{coker}/r$ since coker is finite order, from which it follows that $\#M/rM = \#\prod \Lambda/(f_i^{n_i}, r)$.

In any case, it remains to be shown that

$$\#\prod \Lambda/(f_i^{n_i}, r) = \#\Lambda\left(\prod f_i^{n_i}, r\right).$$

Again, if $f_i \mid r$ for some i , then both sides are infinite. Otherwise, multiplying the exact sequence

$$0 \rightarrow \Lambda/(f) \rightarrow \Lambda/(fg) \rightarrow \Lambda/(g) \rightarrow 0$$

by r and applying the snake lemma gives

$$0 \rightarrow \Lambda/(f)[r] \rightarrow \Lambda/(fg)[r] \rightarrow \Lambda/(g)[r] \rightarrow \Lambda/(f, r) \rightarrow \Lambda/(fg, r) \rightarrow \Lambda/(g, r) \rightarrow 0.$$

If r is prime to g , then $\Lambda(g)[r] = 0$ and $\#\Lambda/(fg) = \#\Lambda/(f) \cdot \#\Lambda/(g)$. Thus, the result follows by inducting on the number of f_i . $\square$

4.2 Classical formulation of the Main Conjecture

Iwasawa's theorem discussed above makes it clear that the Λ -module X_∞ carries the deep arithmetic information of a family of cyclotomic class groups. We may recall from section 2 that $\text{Cl}(\mathbb{Q}(\mu_p))$ admits a finer decomposition in terms of the isotypic components for the action of $\text{Gal}(\mathbb{Q}(\mu_p)/\mathbb{Q})$. One is then naturally led to ask if there is some refinement of X_∞ that captures the information of this decomposition.

Fixing N and letting K , K_n , etc. be as before, suppose that $p^2 \nmid N$. Then $p \nmid [K : \mathbb{Q}]$ and $K \cap \mathbb{Q}_\infty = \mathbb{Q}$, and so we have

$$\text{Gal}(K_\infty/\mathbb{Q}) = \text{Gal}(K_\infty/K) \times \text{Gal}(K/\mathbb{Q}).$$

In particular, letting $X_\infty = \text{Gal}(L_\infty/K_\infty)$ as before, we obtain an action of $\text{Gal}(K/\mathbb{Q})$ on X_∞ . Let ψ be an odd primitive Dirichlet character of conductor N ; in other words, let ψ be a character of $\text{Gal}(K/\mathbb{Q})$ sending complex conjugation to -1 .

Let $\mathcal{O}_\psi = \mathbb{Z}_p[\psi]$ be $\mathbb{Z}_p$ adjoined with the values of ψ . Then we may put

$$X_\infty^\psi := X_\infty \otimes_{\mathbb{Z}_p[\text{Gal}(K/\mathbb{Q})]} \mathcal{O}_\psi,$$

where we let $\text{Gal}(K/\mathbb{Q})$ act on $\mathcal{O}_\psi$ by ψ . Then X_∞^ψ is essentially the ψ -isotypic component of X_∞ , and since Iwasawa proved that X_∞ is a finite torsion Λ -module, it follows that X_∞^ψ is a finite torsion $\Lambda_{\mathcal{O}_\psi}$ -module.

The Herbrand-Ribet theorem establishes a relationship between the ψ -part of $\text{Cl}(\mathbb{Q}(\mu_p))$ and the vanishing mod p of a certain L -value. Iwasawa's conjecture can be seen as a related, much stronger statement, relating the ψ -parts of a large family of class groups to a family of L -values.

Theorem 5 (Main Conjecture, classical statement). *Let $g_\psi \in \mathcal{O}_\psi[[T]]$ be the p -adic L -function associated to ψ as in section 3. Suppose that $p \nmid [\mathbb{Q}(\mu_N) : \mathbb{Q}]$. Then*

$$\text{Ch}(X_\infty^\psi) = (g_\psi).$$

Rather than focusing on this statement, we will instead work on a version using language more in line with modern Iwasawa theory. Our more modern formulation will have the benefit of addressing character with conductor divisible by higher powers of p .

4.3 The Selmer group of a Galois character

Now we will give a more modern formulation of the Main Conjecture in terms of Selmer groups, which are cohomological invariants that may be attached to Dirichlet characters (or Galois representations in general). This formulation is more aligned with modern treatments and applications of Iwasawa theory, though Λ -modules and the crucial idea of p -adic interpolation still remain the core of the methodology.

First we will define a notion of a p -adic family of Galois characters. It is to such families that we ultimately wish to attach cohomological invariants.

Let $\Gamma = \text{Gal}(\mathbb{Q}_\infty/\mathbb{Q})$, and fix a topological generator γ of Γ . As before, we let $\Lambda = \mathbb{Z}_p[[\Gamma]]$. Let $\Psi : G_{\mathbb{Q}} \rightarrow \Lambda^\times$ be the projection $G_{\mathbb{Q}} \rightarrow \Gamma$ composed with the natural inclusion $\Gamma \hookrightarrow \Lambda^\times$.

We may observe that the characters of Λ may be identified with the open unit disk, in the sense that $\Lambda \cong \mathbb{Z}_p[[T]]$, and so for any p -adic field L , we may identify $\text{hom}_{cts}(\Lambda, L)$ with the unit disk in L by $\phi \mapsto \phi(\gamma) = \phi(1+T)$. The upshot of this is that as we let $\phi \in \text{hom}_{cts}(\Lambda, L)$ vary, the compositions $\phi \circ \Psi$ form a “ p -adic family” of Galois characters. In this sense we often simply refer to Ψ as a family of characters. Moreover, for any fixed Galois character ψ , we have the family of characters $\psi\Psi$; the case of ψ a Dirichlet character will be the key families we will study.

So let ψ be a primitive Dirichlet character of conductor N . Write $F_\psi = \mathbb{Q}_p[\psi]$ and $\mathcal{O}_\psi = \mathbb{Z}_p[\psi]$, and as before let $\Lambda_\psi = \Lambda \otimes_{\mathbb{Z}_p} \mathcal{O}_\psi$. We define the *Selmer group* attached to ψ to be the Galois cohomology group

$$\text{Sel}(\psi) := H_{unr}^1(\mathbb{Q}, F_\psi/\mathcal{O}_\psi(\psi)),$$

where H_{unr}^1 denotes the classes in H^1 which are everywhere unramified; that is, those classes in the kernel of the restriction homomorphisms

$$H^1(\mathbb{Q}, F_\psi/\mathcal{O}_\psi(\psi)) \xrightarrow{res} H^1(I_v, F_\psi/\mathcal{O}_\psi(\psi))$$

for all places v of $\mathbb{Q}$. Slightly more generally, we define for the Tate-twists of ψ

$$\text{Sel}(\psi\chi^k) := H_{unr}^1(\mathbb{Q}, F_\psi/\mathcal{O}_\psi(\psi\chi^k)),$$

where χ is the p -adic cyclotomic character. Very roughly speaking, we expect $\text{Sel}(\psi)$ to contain similar arithmetic information to the “ ψ -part” of $\text{Cl}(\mathbb{Q}(\mu_N))$. A coarse way to make this more precise is the following.

Proposition 4. *Let ψ be a primitive Dirichlet character of conductor N , and assume $p \nmid [\mathbb{Q}(\mu_N) : \mathbb{Q}]$. Then we have*

$$\#\text{Sel}(\psi) = \#(\text{Cl}(\mathbb{Q}(\mu_N)) \otimes_{\mathbb{Z}_p} \mathcal{O}_\psi)^\psi.$$

Proof. We may write

$$H_{unr}^1(\mathbb{Q}, F_\psi/\mathcal{O}_\psi(\psi)) = (H_{unr}^1(\mathbb{Q}(\mu_N), F_\psi/\mathcal{O}_\psi(\psi)))^{\text{Gal}(\mathbb{Q}(\mu_N)/\mathbb{Q})} = \text{hom}_{\text{Gal}(\mathbb{Q}(\mu_N)/\mathbb{Q}), unr}(G_{\mathbb{Q}(\mu_N)}, F_\psi/\mathcal{O}_\psi(\psi)).$$

Note that we have equalities of unramified parts due to the naturality of restriction maps. Any map on the right hand side must factor through the maximal unramified abelian extension of $\mathbb{Q}(\mu_N)$, whose Galois group over $\mathbb{Q}(\mu_N)$ is identified $\text{Gal}(\mathbb{Q}(\mu_N)/\mathbb{Q})$ -equivariantly with $\text{Cl}(\mathbb{Q}(\mu_N))$ by class field theory. Thus we have

$$\begin{aligned} \text{hom}_{\text{Gal}(\mathbb{Q}(\mu_N)/\mathbb{Q}), unr}(G_{\mathbb{Q}(\mu_N)}, F_\psi/\mathcal{O}_\psi(\psi)) &= \text{hom}_{\text{Gal}(\mathbb{Q}(\mu_N)/\mathbb{Q})}(\text{Cl}(\mathbb{Q}(\mu_N)), F_\psi/\mathcal{O}_\psi(\psi)) \\ &= \text{hom}_{\text{Gal}(\mathbb{Q}(\mu_N)/\mathbb{Q})}(\text{Cl}(\mathbb{Q}(\mu_N))[p^\infty], F_\psi/\mathcal{O}_\psi(\psi)), \end{aligned}$$

where the second equality follows since $F_\psi/\mathcal{O}_\psi$ is p -power torsion.

Any map on the right uniquely extends to an $\mathcal{O}_\psi$ -map

$$\text{Cl}(\mathbb{Q}(\mu_N))[p^\infty] \otimes_{\mathbb{Z}_p} \mathcal{O}_\psi \rightarrow F_\psi/\mathcal{O}_\psi(\psi),$$

and each such map is obtained this way. Now since $p \nmid N^2$, $\text{Gal}(\mathbb{Q}(\mu_N)/\mathbb{Q})$ acts semisimply on the left, so any map to the right must factor through the ψ -isotypic component of the left. So now we just need to count the number of $\mathcal{O}_\psi$ -maps

$$\left(\text{Cl}(\mathbb{Q}(\mu_N))[p^\infty] \otimes_{\mathbb{Z}_p} \mathcal{O}_\psi \right)^\psi \rightarrow F_\psi / \mathcal{O}_\psi,$$

which is just $\#(\text{Cl}(\mathbb{Q}(\mu_N))[p^\infty] \otimes_{\mathbb{Z}_p} \mathcal{O}_\psi)^\psi$. $\square$

In this way, we see that the Selmer groups we've defined can be thought of as generalized class groups which carry information about the 1-dimensional Galois representation ψ . Moreover, there is an obvious analogy between these Selmer groups and the p -Selmer groups of elliptic curves, which should carry some information about the p -adic Galois representation attached to the elliptic curve. Both of these notions of Selmer groups should be thought of as incarnations of a more general cohomological invariant one can attach to more complicated Galois representations and geometric objects.

The classical formulation of the Main Conjecture related the p -adic L -function to a module given by piecing together a family of class groups. Our reformulation will be of the same spirit; we will now define an object which serves as an interpolation of a family of Selmer groups, and then relate this object to the p -adic L -function.

For the rest of this article, for a topological $\mathbb{Z}_p$ -module M we let M^* denote the Pontryagin dual:

$$M^* = \text{hom}_{cts}(M, \mathbb{Q}/\mathbb{Z}) = \text{hom}_{cts}(M, \mathbb{Q}_p/\mathbb{Z}_p).$$

Note further that if M is in fact a continuous $\mathcal{O}_\psi$ -module, then we also have

$$M^* = \text{hom}_{cts, \mathcal{O}_\psi}(M, F_\psi / \mathcal{O}_\psi).$$

We may define

$$\text{Sel}_\infty(\psi) := H_{unr}^1(\mathbb{Q}, A_\psi^*(\psi\Psi^{-1})).$$

Note that A_ψ^* is a Λ -module by putting $(\lambda \cdot \varphi)(x) = \varphi(\lambda x)$ for $\lambda \in A_\psi$ and $\varphi \in A_\psi^*$; this is the diagonal action resulting from giving $F_\psi / \mathcal{O}_\psi$ the trivial Λ action. Then $\text{Sel}_\infty(\psi)$ inherits a Λ_ψ structure in the natural way.

Proposition 5 (Control Theorem). *Let $k \geq 0$, and for ζ a p -power root of unity let ψ_ζ be the character of $G_\mathbb{Q}$ factoring through Γ and sending $\gamma \mapsto \zeta$. Suppose that ψ is not p -power order. Then we have*

$$\text{Sel}(\psi\psi_\zeta^{-1}\omega^k\chi^{-k}) \xrightarrow{\sim} (\text{Sel}_\infty(\psi) \otimes_{\mathcal{O}_\psi} \mathcal{O}_\psi[\zeta])[\gamma - \zeta\chi^k(\gamma)],$$

where on the right we mean the $(\gamma - \zeta\chi^k(\zeta))$ -torsion coming from the Λ -module structure.

In this way, we see that $\text{Sel}_\infty(\psi)$ interpolates a large family of Selmer groups into a single Λ_ψ -module.

Proof. For convenience abbreviate $F = \mathbb{Q}_p[\psi, \zeta]$, $\mathcal{O} = \mathbb{Z}_p[\psi, \zeta]$, and $A := A_\psi[\zeta]$. Also, we write $\Psi_{k, \zeta}$ for the character $\psi_\zeta\omega^{-k}\chi^k$; note that $\Psi_{k, \zeta}$ is the composition of Ψ with the map sending $\gamma \mapsto \zeta\chi^k(\gamma)$. Observe that since $\mathcal{O}_\psi[\zeta]$ is a flat $\mathcal{O}_\psi$ -module, we have

$$\text{Sel}_\infty(\psi) \otimes_{\mathcal{O}_\psi} \mathcal{O}_\psi[\zeta] = H_{unr}^1(\mathbb{Q}, A_\psi^*(\psi\Psi^{-1}) \otimes_{\mathcal{O}_\psi} \mathcal{O}_\psi[\zeta]) = H_{unr}^1(\mathbb{Q}, A^*(\psi\Psi^{-1})).$$

We will work on the latter cohomology group. We may define a map

$$\phi : A(\psi\Psi^{-1}) \rightarrow \mathcal{O}(\psi\Psi_{k, \zeta}^{-1})$$

of $\mathcal{O}[G_\mathbb{Q}]$ -algebras by sending $\gamma \mapsto \zeta\chi^k(\gamma)$ and extending continuously and $\mathcal{O}$ -linearly. Let $K := \ker \phi$. Taking Pontryagin duals we obtain the exact sequence

$$0 \rightarrow F/\mathcal{O}(\psi\Psi_{k, \zeta}^{-1}) \rightarrow A^*(\psi\Psi^{-1}) \rightarrow K^* \rightarrow 0$$

of Galois modules. Consider the relevant long exact sequence on cohomology:

$$\cdots \rightarrow H^0(\mathbb{Q}, K^*) \rightarrow H^1(\mathbb{Q}, F/\mathcal{O}(\psi\Psi_{k,\zeta}^{-1})) \xrightarrow{\eta} H^1(\mathbb{Q}, A^*(\psi\Psi^{-1})) \rightarrow H^1(\mathbb{Q}, K^*) \rightarrow \cdots$$

The map η will turn out to give our desired isomorphism when suitably restricted.

First, we show η is injective by showing that $H^0(\mathbb{Q}, K^*) = 0$. Indeed, suppose that $\varphi \in K^*$ were $G_{\mathbb{Q}}$ -invariant. Then for $x \in K$ and $\sigma \in G_{\mathbb{Q}}$, we have

$$(\sigma\varphi)(x) = \varphi(\psi\Psi^{-1}(\sigma)x) = \psi(\sigma) \cdot \varphi(\Psi^{-1}(\sigma)x) = \varphi(x).$$

Thus, to show $\varphi = 0$, it suffices to find $\sigma \in G_{\mathbb{Q}}$ with $\Psi(\sigma) = 1$ and $\psi(\sigma) \not\equiv 1 \pmod{\pi}$, where π is a uniformizer for $\mathcal{O}$; for then we have $(\psi(\sigma) - 1)\varphi(x) = 0$ for all $x \in K$, and $\psi(\sigma) - 1$ is a unit of $\mathcal{O}$.

Suppose that $p^e \parallel N$ with $e \geq 1$, and that ψ takes some element σ' of $\text{Gal}(\mathbb{Q}(\mu_N)/\mathbb{Q}(\mu_{N/p^e}))$ to a root of unity which is not p -power torsion. Replacing σ' with $\sigma'^{p^{e-1}}$, we may assume that $\psi(\sigma')$ is a nontrivial root of unity of prime to p order (and in particular is not 1 modulo π). Then since $\text{Gal}(\mathbb{Q}(\mu_N)/\mathbb{Q}(\mu_{N/p^e}))$ is a direct summand of $\text{Gal}(\mathbb{Q}(\mu_N)/\mathbb{Q})$ corresponding to the subfield $\mathbb{Q}(\mu_{p^e})$, it follows that if σ is a lift of σ' to $G_{\mathbb{Q}}$, then σ acts trivially on $\mathbb{Q}_{\infty}$, i.e. has trivial image in Γ .

Otherwise, we may assume that for some $\ell \mid N$ not equal to p , say with $\ell^e \parallel N$, that ψ maps some element of $\text{Gal}(\mathbb{Q}(\mu_N)/\mathbb{Q}(\mu_{\ell^e}))$ to a non- p -power root of unity. Then any lift of this element to an element of $G_{\mathbb{Q}}$ maps to 1 in Γ , as desired. These two cases are exhaustive since we assume ψ is not p -power order, so $H^0(\mathbb{Q}, K^*)$ is trivial as desired, and η is injective.

Now we will show that η is an isomorphism onto the $(\gamma - \zeta\chi^k(\gamma))$ -torsion of its target; then, since η is injective, the naturality of the restriction maps for inertia guarantee that η will give an isomorphism on the corresponding unramified parts of cohomology, as desired.

To see that η has image in $(\gamma - \zeta\chi^k(\gamma))$ -torsion, merely observe that for $d \in Z^1(G_{\mathbb{Q}}, F/\mathcal{O}(\psi\Psi_{k,\zeta}^{-1}))$ a cocycle and $c \in Z^1(G_{\mathbb{Q}}, A^*(\psi\Psi^{-1}))$ the image of d , we have

$$(\gamma \cdot c(\sigma))(x) = c(\sigma)(\gamma x) = d(\sigma)(\phi(\gamma x)) = d(\sigma)(\zeta\chi^k(\gamma) \cdot \phi(x)) = \zeta\chi^k c(\sigma)(x).$$

Now we need surjectivity. For this, it suffices to show that

$$H^1(\mathbb{Q}, A^*(\psi\Psi^{-1}))[\gamma - \zeta\chi^k(\gamma)] \rightarrow H^1(\mathbb{Q}, K^*)$$

is the zero map. For any $c \in Z^1(G_{\mathbb{Q}}, A^*(\psi\Psi^{-1}))$ representing a class on the left hand side, we have that

$$(\gamma - \zeta\chi^k(\gamma))c$$

is a coboundary. Then noting that K is the ideal of A generated by $\gamma - \zeta\chi^k(\gamma)$, we may write $(\gamma - \zeta\chi^k(\gamma))x$ for a generic element of K . Then

$$c(\sigma)((\gamma - \zeta\chi^k(\gamma))x) - (\gamma - \zeta\chi^k(\gamma))c(\sigma)(x) = \sigma y(x) - y(x) \text{ for some } y \in A^*.$$

In particular, regarding y as an element of K^* , we see that c maps to a coboundary in $H^1(\mathbb{Q}, K^*)$. Thus, η surjects onto $H^1(\mathbb{Q}, A^*(\psi\Psi^{-1}))[\gamma - \zeta\chi^k(\gamma)]$, as desired. $\square$

4.4 The Main Conjecture in terms of Selmer groups

In this section we will use the notation of both Sections 4.1 and 4.2. We define the following objects attached to a primitive Dirichlet character ψ :

$$X(\psi\chi^k) := \text{Sel}(\psi\chi^k)^*, \quad X_{\infty}(\psi) := \text{Sel}_{\infty}(\psi)^* = \text{hom}_{\Lambda_{\psi}}(\text{Sel}_{\infty}(\psi), \Lambda_{\psi}^*).$$

Note of course that $X_{\infty}(\psi)$ carries a Λ_{ψ} -module structure. Note also that as an immediate corollary of the Control Theorem, we obtain isomorphisms

$$(X_\infty(\psi) \otimes_{\mathcal{O}_\psi} \mathcal{O}_\psi[\zeta]) / (\gamma - \zeta \chi^k(\gamma)) \xrightarrow{\sim} X(\psi \psi_\zeta^{-1} \omega^k \chi^{-k})$$

for appropriate ψ , k , and ζ .

In this way, we see that $X_\infty(\psi)$ interpolates a large family of arithmetic data for Galois characters, similarly to how the Galois group X_∞ interpolated a family of ideal class groups. In reformulating the Main Conjecture, we will analogously relate the Λ_ψ -module structure of $X_\infty(\psi)$ to the p -adic L -function.

Identifying $\gamma \in \Lambda_\psi$ with $1+T \in \mathcal{O}_\psi[[T]]$, by Weierstrass preparation we may write any $f \in \Lambda_\psi$ as $f = \pi^{\mu(f)} Q u$, where π is a uniformizer for $\mathcal{O}_\psi$, Q is a distinguished polynomial of degree $\lambda(f)$, and u is a unit of $\mathcal{O}_\psi[[T]]$. The values $\mu(f)$ and $\lambda(f)$ depend only on f . In particular, for a finite Λ -module M , if M is torsion then $Ch(M) \neq 0$, and so the values $\lambda(f)$ and $\mu(f)$ for a generator f of $Ch(M)$ depend only on M .

The following theorem is the first key step toward relating Selmer groups to L -values.

Theorem 6 (Iwasawa). *$X_\infty(\psi)$ is a finite torsion Λ_ψ -module. Let $f_\psi = Ch(X_\infty(\psi))$. Let $\mathcal{O}$ be the integer ring of a finite extension of $\mathbb{Q}_p$ containing all values of all odd characters in $\text{Gal}(\overline{K}/\mathbb{Q})$, where $K = \mathbb{Q}(\mu_N)$ as before. Then if f_ψ is a generator for $Ch(X_\infty(\psi))$, we have*

$$\sum_{\substack{\psi \in \widehat{\text{Gal}(\overline{K}/\mathbb{Q})} \\ \psi \text{ odd}}} \mu(f_\psi) = \sum_{\substack{\psi \in \widehat{\text{Gal}(\overline{K}/\mathbb{Q})} \\ \psi \text{ odd}}} \mu(g_\psi), \quad \sum_{\substack{\psi \in \widehat{\text{Gal}(\overline{K}/\mathbb{Q})} \\ \psi \text{ odd}}} \lambda(f_\psi) = \sum_{\substack{\psi \in \widehat{\text{Gal}(\overline{K}/\mathbb{Q})} \\ \psi \text{ odd}}} \lambda(g_\psi),$$

where, as before, g_ψ is the power series in $\mathcal{O}[[T]]$ for the p -adic L -function defined in section 3, and where all μ 's and λ 's are computed in $\mathcal{O}[[T]]$. Moreover, if $p \nmid [K : \mathbb{Q}]$, then $X_\infty(\psi)$ has no finite-order submodules.

Proof. We follow the proof in [Sk09]. By Shapiro's lemma, we have an isomorphism

$$H_{unr}^1(K, \Lambda_{\mathcal{O}}^*(\Psi^{-1})) = H_{unr}^1(\mathbb{Q}, \text{hom}_{\mathcal{O}}(\mathcal{O}[\text{Gal}(\overline{K}/\mathbb{Q})], \Lambda_{\mathcal{O}}^*(\Psi^{-1}))).$$

Moreover, comparing the $\Lambda_{\mathcal{O}}$ -actions on both sides, it is easy to see that this is a $\Lambda_{\mathcal{O}}$ -isomorphism. Now consider the map

$$\phi : \prod_{\psi \in \widehat{\text{Gal}(\overline{K}/\mathbb{Q})}} \mathcal{O}(\psi) \rightarrow \text{hom}_{\mathcal{O}}(\mathcal{O}[\text{Gal}(\overline{K}/\mathbb{Q})], \mathcal{O})$$

given by $(a_\psi) \mapsto (g \mapsto \sum a_\psi \psi^{-1}(g))$. ϕ is injective and $\mathcal{O}[G_{\mathbb{Q}}]$ equivariant. A given homomorphism φ is in the image of ϕ iff there are $a_{\psi_1}, \dots, a_{\psi_n}$ so that

$$\begin{pmatrix} \psi_1^{-1}(g_1) & \cdots & \psi_n^{-1}(g_1) \\ \vdots & & \vdots \\ \psi_1^{-1}(g_n) & \cdots & \psi_n^{-1}(g_n) \end{pmatrix} \begin{pmatrix} a_{\psi_1} \\ \vdots \\ a_{\psi_n} \end{pmatrix} = \begin{pmatrix} \varphi(g_1) \\ \vdots \\ \varphi(g_n) \end{pmatrix}.$$

From this it is clear that our map has finite cokernel, controlled by the determinant of the above matrix. In particular, if $p \nmid [K : \mathbb{Q}]$, it can be checked that this matrix is $\mathcal{O}$ -invertible, so ϕ is in fact an isomorphism in this case.

In any case, tensoring with $\Lambda_{\mathcal{O}}^*(\Psi^{-1})$, we get an induced map on cohomology

$$\phi_0 : \prod_{\psi \in \widehat{\text{Gal}(\overline{K}/\mathbb{Q})}} H_{unr}^1(\mathbb{Q}, \Lambda_{\mathcal{O}}^*(\psi \Psi^{-1})) \rightarrow H_{unr}^1(\mathbb{Q}, \text{hom}_{\mathcal{O}}(\mathcal{O}[\text{Gal}(\overline{K}/\mathbb{Q})], \Lambda_{\mathcal{O}}^*(\Psi^{-1}))) = H_{unr}^1(K, \Lambda_{\mathcal{O}}^*(\Psi^{-1}))$$

with finite kernel and cokernel (which is an isomorphism if $p \nmid [K : \mathbb{Q}]$). Note that ϕ is equivariant for the action of $\text{Gal}(\overline{K}/\mathbb{Q})$ on its source and target, and that this $\text{Gal}(\overline{K}/\mathbb{Q})$ action propagates down to the source and target of ϕ_0 , which is also equivariant for this action. Thus, we may consider the restriction of ϕ_0 to the subgroups where $\text{Gal}(\overline{K}/K^+)$ acts non-trivially (where K^+ is the maximal totally real subfield of K), giving a map

$$\prod_{\substack{\psi \in \widehat{\text{Gal}(K/\mathbb{Q})} \\ \psi \text{ odd}}} H_{unr}^1(\mathbb{Q}, \Lambda_{\mathcal{O}}^*(\psi\Psi^{-1})) \rightarrow H_{unr}^1(K, \Lambda_{\mathcal{O}}^*(\Psi^{-1}))^{-},$$

where the superscript $\cdot^{-}$ denotes the nontrivial $\text{Gal}(K/K^+)$ -isotypic piece.

The left-hand side of the above is just a product of the Selmer groups $\text{Sel}_{\infty}(\psi) \otimes_{\mathcal{O}_{\psi}} \mathcal{O}$; letting $X_{\infty}(K)^{-}$ be the Pontryagin dual of the right side, taking duals on both sides gives a map

$$X_{\infty}(K)^{-} \rightarrow \prod_{\substack{\psi \in \widehat{\text{Gal}(K/\mathbb{Q})} \\ \psi \text{ odd}}} X_{\infty}(\psi) \otimes_{\mathcal{O}_{\psi}} \mathcal{O}$$

with finite order kernel and cokernel (which is an isomorphism if $p \nmid [K : \mathbb{Q}]$).

The restriction of Ψ to G_K factors through the subgroup $\Gamma_K := \text{Gal}(K_{\infty}/K)$ (which is identified with the subgroup $\text{Gal}(\mathbb{Q}_{\infty}/(\mathbb{Q}_{\infty} \cap K))$ of Γ). Thus, we have

$$H_{unr}^1(K, \Lambda_{\mathcal{O}}^*(\Psi^{-1})) \xrightarrow{\sim} \text{hom}_{\Gamma_K, unr}(G_{K_{\infty}}, \Lambda_{\mathcal{O}}^*(\Psi^{-1})).$$

Moreover, since $\Lambda_{\mathcal{O}}^*$ is ind- p , any map on the right must factor through the maximal unramified abelian pro- p extension of K_{∞} , which is L_{∞} (as defined in section 4.1). Thus, we in fact have

$$H_{unr}^1(K, \Lambda_{\mathcal{O}}^*(\Psi^{-1})) \xrightarrow{\sim} \text{hom}_{\Gamma_K}(X_{\infty}, \Lambda_{\mathcal{O}}^*(\Psi^{-1})).$$

The Pontryagin dual of the right hand side is

$$\begin{aligned} & \text{hom}_{cts}(\text{hom}_{\Gamma_K}(X_{\infty}, \Lambda_{\mathcal{O}}^*(\Psi^{-1})), \mathbb{Q}_p/\mathbb{Z}_p) \\ & \xrightarrow{\sim} \text{hom}_{\Lambda_{\mathcal{O}}}(\text{hom}_{\Gamma_K}(X_{\infty}, \Lambda_{\mathcal{O}}^*(\Psi^{-1})), \Lambda_{\mathcal{O}}^*) \\ & \xrightarrow{\sim} X_{\infty}^{\iota} \otimes_{\mathbb{Z}_p[[\Gamma_K]]} \Lambda_{\mathcal{O}}, \end{aligned}$$

where isomorphisms are of $\Lambda_{\mathcal{O}}$ -modules, and the ι indicates that the usual Γ_K -action on X_{∞} has been inverted; this is necessary due to the Ψ^{-1} twist on the inner $\Lambda_{\mathcal{O}}^*$. This nets us a $\Lambda_{\mathcal{O}}$ -isomorphism

$$X_{\infty}^{-, \iota} \otimes_{\mathbb{Z}_p[[\Gamma_K]]} \Lambda_{\mathcal{O}} \xrightarrow{\sim} X_{\infty}(K)^{-},$$

where, as before, the $-$ indicates the piece on which $\text{Gal}(K/K^+)$ acts nontrivially (note the map we obtained is equivariant for the $\text{Gal}(K/\mathbb{Q})$ -action on each side). For convenience, denote the left side by $X_{\infty, \mathcal{O}}^{-, \iota}$.

So far, we have produced a map from $X_{\infty, \mathcal{O}}^{-, \iota}$ to the product of the $X_{\infty}(\psi)$'s (tensored with $\mathcal{O}$) with finite kernel and cokernel (which is an isomorphism if $p \nmid [K : \mathbb{Q}]$). Iwasawa proved that X_{∞} is a finite, torsion Λ -module with no finite submodules, from which we may deduce the first part of the theorem for each $X_{\infty}(\psi)$. Since $X_{\infty}(\psi)$ is now confirmed to be finitely generated, let f_{ψ} be a generator of its characteristic ideal.

In Iwasawa's study of X_{∞} , he gave a submodule Y of X_{∞}^{-} with the property that, for $n \geq n_0$, if we put

$$\nu_n := \frac{\gamma^{p^n} - 1}{\gamma^{p^{n_0}} - 1},$$

then $X_{\infty}/\nu_n Y \simeq \text{Gal}(L_n/K_n)^{-}$. In particular, Y has finite index in X , so $Y \otimes_{\mathbb{Z}_p[[\Gamma_K]]} \Lambda_{\mathcal{O}}$ has characteristic ideal generated by the product of the f_{ψ} 's (i.e. has characteristic ideal $Ch(X_{\infty}(K)^{-})$). We may then write

$$\begin{aligned} \# \Lambda_{\mathcal{O}} / (Ch(X_{\infty}(K)^{-}, \nu_n)) &= \# \left((Y \otimes_{\mathbb{Z}_p[[\Gamma_K]]} \Lambda_{\mathcal{O}}) / \nu_n \right) \\ &= (\# Y / \nu_n)^{[K \cap \mathbb{Q}_{\infty} : \mathbb{Q}][\mathcal{O} : \mathbb{Z}_p]} \\ &= \left(\frac{\#(X_{\infty}^{-} / \nu_n Y)}{\#(X_{\infty}^{-} / Y)} \right)^{[K \cap \mathbb{Q}_{\infty} : \mathbb{Q}][\mathcal{O} : \mathbb{Z}_p]} \\ &= (h_n / h_{n_0})^{[K \cap \mathbb{Q}_{\infty} : \mathbb{Q}][\mathcal{O} : \mathbb{Z}_p]}, \end{aligned}$$

where the first equality follows from Lemma 3, and

$$h_n = \#\text{Gal}(L_n/K_n)^- = \#Cl(K_n)[p^\infty]^+.$$

Now let's work on the analytic side. Let g_K be the product of the g_ψ for $\psi \in \widehat{\text{Gal}(K/\mathbb{Q})}$ odd. Observing that the injection

$$\Lambda_{\mathcal{O}[\mu_{p^n}]}(\nu_n) \hookrightarrow \prod_{\substack{\zeta^{p^n}=1 \\ \zeta^{p^{n_0}} \neq 1}} \Lambda_{\mathcal{O}[\mu_{p^n}]}(\gamma - \zeta)$$

has finite cokernel, the first line of the following again follows from Lemma 3:

$$\begin{aligned} \#\Lambda_{\mathcal{O}[\mu_{p^n}]}(g_K, \nu_n) &= \prod_{\substack{\zeta^{p^n}=1 \\ \zeta^{p^{n_0}} \neq 1}} \#\Lambda_{\mathcal{O}[\mu_{p^n}]}(g_K, \gamma - \zeta) \\ &= \prod_{\substack{\zeta^{p^n}=1 \\ \zeta^{p^{n_0}} \neq 1}} \#\mathcal{O}[\mu_{p^n}]/(\phi_{0,\zeta}(g_K)) \\ &= \#\mathcal{O}[\mu_{p^n}]/\left(\prod_{\substack{\zeta^{p^n}=1 \\ \zeta^{p^{n_0}} \neq 1}} (\phi_{0,\zeta}(g_K))\right). \end{aligned}$$

Now we compute the product of the $\phi_{0,\zeta}(g_K)$'s. We may choose n_0 large enough so that $K \cap \mathbb{Q}_\infty \subset K_{n_0}$. Then because of the interpolation property of the g_ψ , we have

$$\phi_{0,\zeta}(g_K) = \prod_{\substack{\psi \in \widehat{\text{Gal}(K/\mathbb{Q})} \\ \psi \text{ odd}}} L^{\{p\}}(0, \psi\psi_\zeta).$$

So for $n \geq n_0$ we may then compute

$$\begin{aligned} \prod_{\zeta^{p^n}=1} \phi_{0,\zeta}(g_K) &= \prod_{\zeta^{p^n}=1} \prod_{\substack{\psi \in \widehat{\text{Gal}(K/\mathbb{Q})} \\ \psi \text{ odd}}} L^{\{p\}}(0, \psi\psi_\zeta) \\ &= \prod_{\substack{\psi' \in \widehat{\text{Gal}(\mathbb{Q}_n/\mathbb{Q})} \\ \psi' \text{ odd}}} \prod_{\substack{\psi \in \widehat{\text{Gal}(K/\mathbb{Q})} \\ \psi \text{ odd}}} L^{\{p\}}(0, \psi\psi') \\ &= \prod_{\substack{\psi \in \widehat{\text{Gal}(K_n/\mathbb{Q})} \\ \psi \text{ odd}}} L^{\{p\}}(0, \psi)^{[K \cap \mathbb{Q}_\infty : \mathbb{Q}]}. \end{aligned}$$

Note that the p factors can only be nontrivial when the given character ψ factors through $\text{Gal}(K/\mathbb{Q})$. Thus, by the analytic class number formula, we have

$$\prod_{\substack{\zeta^{p^n}=1 \\ \zeta^{p^{n_0}} \neq 1}} (\phi_{0,\zeta}(g_K)) = (h_n/h_{n_0})^{[K \cap \mathbb{Q}_\infty : \mathbb{Q}]}$$

Since h_n and h_0 are p th powers, it follows that

$$\#\Lambda_{\mathcal{O}[\mu_{p^n}]} \left(\prod_{\substack{\psi \in \widehat{\text{Gal}(K/\mathbb{Q})} \\ \psi \text{ odd}}} g_\psi, \nu_n \right) = (h_n/h_{n_0})^{[K \cap \mathbb{Q}_\infty : \mathbb{Q}][\mathcal{O} : \mathbb{Z}_p]} = \#\Lambda_{\mathcal{O}[\mu_{p^n}]} \left(\prod_{\substack{\psi \in \widehat{\text{Gal}(K/\mathbb{Q})} \\ \psi \text{ odd}}} f_\psi, \nu_n \right)$$

All that is left is to relate the quantities on each side to the values of μ and λ for each g_ψ and f_ψ . It can be shown that for $r \in \mathcal{O}$, writing $\#\Lambda_{\mathcal{O}}(r, \nu_n) = p^a$, we have

$$a = \mu(r)fp^{n-n_0} + \lambda(r)ef(n - n_0) + c_n$$

where f is the residue extension degree of $\mathcal{O}$ over $\mathbb{Z}_p$, e is the ramification degree, and c_n is bounded uniformly in n ([Wa97, §13]). Thus, comparing $r = \prod f_\psi$ and $r = \prod g_\psi$ gives the desired result. $\square$

The above theorem is the first evidence of a relationship between the analytically obtained power series g_ψ and the algebraically obtained f_ψ . In parallel with the classical form, it is then natural to formulate the Main Conjecture in this setting in the following way:

Theorem 7 (Main Conjecture, Selmer group form). *Let ψ be an odd primitive Dirichlet character. Then we have*

$$Ch(X_\infty(\psi)) = (g_{\psi^{-1}})$$

This more refined “algebraic = analytic” statement is representative of a more general phenomenon for p -adic Galois representations, which is what motivates discussing the Main Conjecture in this form. We can see now that this form implies the classical form:

Proof (Theorem 7 $\Rightarrow$ Theorem 5). Since we assume that $p \nmid [K : \mathbb{Q}]$, we have the isomorphism from the proof of Theorem 6:

$$X_\infty^{-,\iota} \otimes_{\mathbb{Z}_p[[\Gamma_K]]} \Lambda_{\mathcal{O}} \xrightarrow{\sim} \prod_{\substack{\psi \in \widehat{\text{Gal}(K/\mathbb{Q})} \\ \psi \text{ odd}}} X_\infty(\psi) \otimes_{\mathcal{O}_\psi} \mathcal{O}$$

Since this isomorphism is $\text{Gal}(K/\mathbb{Q})$ -equivariant, we may tensor with $\mathcal{O}(\psi)$ on both sides, obtaining

$$X_\infty^{-,\iota} \otimes_{\mathbb{Z}_p[[\Gamma_K]]} \Lambda_{\mathcal{O}} \otimes_{\mathcal{O}[\text{Gal}(K/\mathbb{Q})]} \mathcal{O}(\psi) \xrightarrow{\sim} X_\infty(\psi) \otimes_{\mathcal{O}_\psi} \mathcal{O}$$

Now note that because the Galois action on the left is inverted, the left side is isomorphic to $X_\infty^{\psi^{-1}}$ tensored up with $\Lambda_{\mathcal{O}}$; then the $\Lambda_{\mathcal{O}}$ characteristic ideals of each side are the same, from which the result follows. $\square$

Note how the above explains the ψ^{-1} in Theorem 7. We defer the proof of Theorem 7 to section 6; first, we must develop an adequate theory of Galois representations to input into our proof.

5 Hida Theory and Galois Representations

Our strategy for proving the Main Conjecture will mirror the proof strategy for Ribet’s converse. However, the statement of the Main Conjecture is already much more sophisticated than Ribet’s converse. Whereas in Ribet’s converse we wanted to relate a class group to an L -value, for Theorem 7 we essentially want to relate a p -adic family of class group-like objects to a p -adic family of L -values. Accordingly, simply using a Galois representation attached to a classical modular form will not be sufficient to relate the two sides; instead, we will need to use a p -adic family of modular forms, and a corresponding p -adic family of Galois representations. We will now make this notion precise through the lens of Hida theory.

5.1 p -adic analytic families of modular forms

Let ψ be a primitive Dirichlet character of conductor Np^r , where $p \nmid N$. Let $\mathbb{I}$ be a finite Λ_ψ -algebra which is a complete local domain. In particular, $\mathbb{I}$ is profinite. We define an $\mathbb{I}$ -adic modular form to be a q -expansion

$$\mathbf{f} = \sum_{n=0}^{\infty} a_n q^n \in \mathbb{I}[[q]]$$

such that for all $\zeta \in \mu_{p^\infty}$ and for all $k > 0$ sufficiently large, for any $\mathcal{O}_\psi$ -algebra map $\phi : \mathbb{I} \rightarrow \overline{\mathbb{Q}}_p$ restricting to $\phi_{k,\zeta}$ on Λ_ψ we have

$$\phi_{k,\zeta}(\mathbf{f}) = \sum_{n=0}^{\infty} \phi_{k,\zeta}(a_n) q^n \in M_{k+1}(\Gamma_0(Np^{\max(r,t+1)}), \psi\psi_{\zeta}\omega^{-k}, \phi(\mathbb{I}))$$

where t is so that ζ is a primitive p^t th root of unity. Moreover, we say that $\mathbf{f}$ is an $\mathbb{I}$ -adic cusp form if all but finitely many specializations are cusp forms.

In the case of $\mathbb{I} = \Lambda_{\psi}$, we have the following interpretation: an element a of Λ_{ψ} is a power series in $\mathcal{O}_{\psi}$, and evaluating $\phi_{k,\zeta}$ is the same as evaluating a at the special point of the unit disk corresponding to $\phi_{k,\zeta}$. Thus, we require that $\mathbf{f}$ be a “modular form” whose coefficients are power series that give genuine modular forms when specialized at particular special points. Because $\mathbb{I}$ -adic modular forms may be interpreted in this way as families of modular forms which vary p -adically, we sometimes refer to $\mathbb{I}$ -adic forms as p -adic families of modular forms, or in the ordinary case (discussed below) more briefly as Hida families.

The most important Hida family for us will be the Eisenstein family for ψ , which will be the family allowing us to generalize the procedure used to prove Ribet’s converse. This family is defined by

$$\mathcal{E}_{\psi} := g_{\psi} + h_{\psi} \sum_{n=1}^{\infty} \left(\sum_{\substack{d|n \\ (d,Np)=1}} \prod_{\ell^e || d} \psi\psi(\text{Frob}_{\ell}^e) \right) q^n$$

To see that $\mathcal{E}_{\psi}$ is a Λ_{ψ} -adic modular form, merely note that

$$\begin{aligned} \phi_{k,\zeta}(\mathcal{E}_{\psi}) &= \phi_{k,\zeta}(g_{\psi}) + \phi_{k,\zeta}(h_{\psi}) \sum_{n=1}^{\infty} \left(\sum_{\substack{d|n \\ (d,Np)=1}} \prod_{\ell^e || d} \psi\psi_{\zeta} \chi^k \omega^{-k}(\text{Frob}_{\ell}^e) \right) q^n \\ &= \psi_{k,\zeta}(h_{\psi}) \left(L^{\{p\}}(1 - k, \psi\psi_{\zeta} \omega^{-k}) + \sum_{n=1}^{\infty} \sum_{\substack{d|n \\ (d,Np)=1}} \psi\psi_{\zeta} \omega^{-k}(d) d^k q^n \right) \\ &= E_{k+1}^{\psi\psi_{\zeta} \omega^{-k}}(z) - p^{k+1} E_{k+1}^{\psi\psi_{\zeta} \omega^{-k}}(pz) \end{aligned}$$

where, as usual, $q = e^{2\pi iz}$, and where $E_{k+1}^{\psi\psi_{\zeta} \omega^{-k}}$ is the Eisenstein series of Nebentypus $\psi\psi_{\zeta} \omega^{-k}$.

Of course, just the Eisenstein series will not be enough for our purposes. The constant term g_{ψ} of $\mathcal{E}_{\psi}$ can be seen as a family of L -values appearing in the constant term. We will attempt to leverage the P -divisibility of g_{ψ} for P prime in Λ_{ψ} (i.e. the vanishing of g_{ψ}) to relate $\mathcal{E}_{\psi}$ to a cusp form modulo P . To achieve this end we will need a satisfactory theory of $\mathbb{I}$ -adic cusp forms.

We will only state the results that are necessary for us; for a more complete introduction to the algebraic theory of $\mathbb{I}$ -adic forms, see [Hi93, §7]. Let $\mathcal{M}(\psi, \mathbb{I})$ and $\mathcal{S}(\psi, \mathbb{I})$ be the modules of $\mathbb{I}$ -adic modular forms and cusp forms respectively of nebentypus ψ .

We can define Hecke operators on $\mathcal{M}(\psi, \mathbb{I})$ and $\mathcal{S}(\psi, \mathbb{I})$ in terms of their action on Fourier expansions. Specifically, for $\mathbf{f} = \sum_{n=0}^{\infty} a_n q^n \in \mathcal{M}(\psi, \mathbb{I})$, we put

$$\begin{aligned} T_{\ell} \mathbf{f} &= \sum_{n=0}^{\infty} (a_{n\ell} + \psi\psi(\text{Frob}_{\ell}) a_{n/\ell}) q^n \quad \text{when } \ell \nmid Np \\ U_{\ell} \mathbf{f} &= \sum_{n=0}^{\infty} a_{n\ell} q^n \quad \text{when } \ell \mid Np \end{aligned}$$

These give well defined actions on $\mathcal{M}(\psi, \mathbb{I})$ because each specialization of $T_{\ell} \mathbf{f}$ is just the usual T_{ℓ} action on the corresponding specialization of $\mathbf{f}$ (and the same for the U_{ℓ}). For the same reason, we also obtain operators on $\mathcal{S}(\psi, \mathbb{I})$.

We say an $\mathbb{I}$ -adic form $\mathbf{f}$ is ordinary (at p) if almost every specialization of $\mathbf{f}$ is ordinary. Write $\mathcal{M}^{\text{ord}}(\psi, \mathbb{I})$ and $\mathcal{S}^{\text{ord}}(\psi, \mathbb{I})$ for the spaces of ordinary $\mathbb{I}$ -adic forms and cuspforms respectively. Recall that we have the following notion of an ordinary projector on classical forms:

Lemma 4. *Let ψ be a Dirichlet character of conductor N . Let $\mathbb{T}$ be the Hecke algebra for $M_k(\Gamma_0(N), \psi, \mathcal{O}_K)$, where K is a finite extension of $\mathbb{Q}_p$. Topologizing $\mathbb{T}$ as a finite $\mathcal{O}_K$ -algebra, the limit*

$$e = \lim_{n \rightarrow \infty} U_p^{n!}$$

exists and is an idempotent of $\mathbb{T}$.

Proof. Since $\mathbb{T}$ is reduced, $\mathbb{T} \otimes_{\mathcal{O}_K} K$ is a product of finite extensions of K . The image of U_p is integral in each of these, so it suffices to show that

$$\lim_{n \rightarrow \infty} \alpha^{n!}$$

exists in $\mathcal{O}_K$ for $\alpha \in \mathcal{O}_K$. For this, let $q = \#(\mathcal{O}_K/\pi)$ for π a uniformizer of $\mathcal{O}_K$. Then if $\alpha \in \mathcal{O}_K^\times$,

$$\alpha^{q^r(q-1)} \equiv 1 \pmod{\pi}$$

Thus, in this case we see that $\lim_{n \rightarrow \infty} \alpha^{n!} = 1$. If $\alpha \in \pi \mathcal{O}_K$, then we see that $\lim_{n \rightarrow \infty} \alpha^{n!} = 0$, which completes the proof. $\square$

The element e of the Hecke algebra constructed above is thus an idempotent whose image is clearly seen to be the ordinary forms in $M_k(\Gamma_0(N), \psi, \mathcal{O}_K)$. As such, we call e the ordinary projector. Note that e also makes sense as an operator on $S_k(\Gamma_0(N), \psi, \mathcal{O}_K)$, and projects onto the ordinary cusp forms.

Letting U_p now be the Hecke operator for $\mathcal{M}(\psi, \mathbb{I})$, it can be shown that the limit $e = \lim U_p^{n!}$ also exists and gives a projection onto $\mathcal{M}^{\text{ord}}(\psi, \mathbb{I})$. Ordinary forms are the nicest class of $\mathbb{I}$ -adic forms, and will be the most relevant for our arguments; for instance, $\mathcal{E}_\psi$ is clearly ordinary. Note that since the Hecke algebra is commutative, and since projection onto ordinary forms has been shown to lie in the Hecke algebra, it makes sense to speak of the “ordinary Hecke algebra” for both $\mathcal{M}^{\text{ord}}(\psi, \mathbb{I})$ and $\mathcal{S}^{\text{ord}}(\psi, \mathbb{I})$. It can be shown that both of these Hecke algebras are reduced, and are free and finite over $\mathbb{I}$. Moreover, we have the following duality relation for ordinary Hecke algebras:

Theorem 8. *Let $\mathbb{T}$ be the Hecke algebra for $\mathcal{M}^{\text{ord}}(\psi, \mathbb{I})$. Then we have*

$$\text{hom}(\mathbb{T}, \mathbb{I}) \simeq \mathcal{M}^{\text{ord}}(\psi, \mathbb{I}), \quad \text{hom}(\mathcal{M}^{\text{ord}}(\psi, \mathbb{I}), \mathbb{I}) \simeq \mathbb{T}$$

For each minimal prime Q of $\mathbb{T}$, the map $\mathbb{T} \rightarrow \mathbb{T}/Q$ gives the eigenvalues of a $(\mathbb{T}/Q)$ -adic ordinary eigenform. In particular, for $\mathbb{I}'$ a sufficiently large finite extension of $\mathbb{I}$, $\mathcal{M}^{\text{ord}}(\psi, \mathbb{I}')$ admits a basis of eigenforms. All of the above statements hold with $\mathcal{M}^{\text{ord}}(\psi, \mathbb{I})$ replaced with $\mathcal{S}^{\text{ord}}(\psi, \mathbb{I})$.

Since $\mathbb{T}$ is reduced, the first part of the above theorem is easily seen to imply the others. The proof is similar to the usual duality relation for the classical Hecke algebra, using the pairing given by evaluating a Hecke operator against a (normalized form) and taking the first Fourier coefficient. For proofs of all of the above facts regarding ordinary $\mathbb{I}$ -adic forms, see [Hi93, §7.3].

5.2 Λ -adic Galois representations

Now we show how a Galois representation can be attached to an $\mathbb{I}$ -adic modular form. Let $\mathbb{K} = \text{Frac}(\mathbb{I})$. Our construction will produce a representation of $G_{\mathbb{Q}}$ on a two-dimensional $\mathbb{K}$ -vector space. This representation will not be continuous for the usual topology on $\mathbb{K}$; since $\mathbb{K}$ is not locally compact, a map $G_{\mathbb{Q}} \rightarrow \text{GL}_2(\mathbb{K})$ that is continuous in the usual sense would have very small image, so this notion is not very useful.

Instead, our representation will be continuous in the following sense: there will exist a finite Galois-stable $\mathbb{I}$ -submodule M of $\mathbb{K}^2$ which $\mathbb{K}$ -spans $\mathbb{K}^2$, so that the map $G_{\mathbb{Q}} \rightarrow \text{End}_{\mathbb{I}}(M)$ is continuous for the $\mathfrak{m}$ -adic topology on M (where $\mathfrak{m}$ is the maximal ideal of $\mathbb{I}$). Since M is a finite $\mathbb{I}$ -module and $\mathbb{I}$ is profinite, $\text{End}_{\mathbb{I}}(M)$ is compact, so this is a more reasonable condition. Note that this is independent of the choice of M , since for any other such M' , we may rescale M' so that M and M' intersect in a lattice, and then it suffices to check the continuity of our representation on $M \cap M'$ by the Artin-Rees lemma. We will simply say that a representation ρ is continuous if ρ is continuous in the above sense.

Our main result will be the following:

Theorem 9 (Hida, Wiles). *Let $\mathbf{f} \in \mathcal{S}^{\text{ord}}(\psi, \mathbb{I})$ be a normalized eigenform with Fourier coefficients $\{a_n\}_{n=1}^{\infty}$. Let $\mathbb{K} = \text{Frac}(\mathbb{I})$. Then there exists a unique Galois representation $\rho : G_{\mathbb{Q}} \rightarrow \text{GL}_2(\mathbb{K})$ that is continuous, absolutely irreducible, unramified away from Np , and satisfies*

$$\det(1 - \rho(\text{Frob}_{\ell})T) = 1 - a_{\ell}T + \psi\Psi(\text{Frob}_{\ell})T^2$$

for $\ell \nmid Np$

It is clear that such a representation must be unique by Chebotarev density and the fact that trace determines the representation in characteristic 0. The key input for our proof of Theorem 7 is the existence of Galois representations attached to classical eigenforms, stated as the following famous theorem of Deligne:

Theorem 10 (Deligne). *Let $f \in S_k(\Gamma_0(N), \chi, L)$ be an eigenform with Fourier coefficients $\{a_n\}_{n=1}^{\infty}$ for some L a finite extension of $\mathbb{Q}_p$. Then there exists a unique Galois representation $\rho : G_{\mathbb{Q}} \rightarrow \text{GL}_2(L)$ that is continuous, absolutely irreducible, unramified away from Np , and satisfies*

$$\det(1 - \rho(\text{Frob}_{\ell})T) = 1 - a_{\ell}T + \chi(\ell)\ell^k T^2$$

for $\ell \nmid Np$.

Suppose for a moment that the representation ρ of Theorem 9 existed. For each prime P of $\mathbb{I}$, we could then consider a residual representation for ρ modulo P . That is, taking some Galois stable lattice M , we may consider the $\text{Frac}(\mathbb{I}/P)$ -valued representation ρ_P of $G_{\mathbb{Q}}$ given on M/PM . ρ_P is also unramified away from Np , and has characteristic polynomials given by the reductions of the characteristic polynomials of ρ modulo P . (Note that since in our setting M may not be free over $\mathbb{I}$, it should be checked that this mod P representation is well-defined; but we will not need this.)

Independently of the existence of ρ , we may ask about the existence of ρ_P for each P . More precisely, for an eigenform $\mathbf{f}$ as in Theorem 9, we call a representation $\rho_P : G_{\mathbb{Q}} \rightarrow \text{GL}_2(\overline{\text{Frac}(\mathbb{I}/P)})$ a residual representation mod P attached to $\mathbf{f}$ if ρ_P is continuous (under the topology on $\text{Frac}(\mathbb{I}/P)$ coming from I), semisimple, unramified away from Np , and has

$$\det(1 - \rho_P(\text{Frob}_{\ell})T) = 1 - a_{\ell}T + \psi\Psi(\text{Frob}_{\ell})T^2$$

in the mod P sense.

Proposition 6. *Let P be the kernel of an $\mathcal{O}_{\psi}$ -algebra map $\phi : \mathbb{I} \rightarrow \overline{\mathbb{Q}_p}$ so that ϕ restricts to $\phi_{k,\zeta}$ on Λ_{ψ} . Then, for k sufficiently large, there exists a residual representation mod P attached to $\mathbf{f}$.*

Proof. For each such map ϕ and for $k \gg 0$, $\phi(\mathbf{f})$ is an eigenform in $S_{k+1}(\Gamma_0(Np^{\max(s,t+1)}), \psi\psi_{\zeta}\omega^{-k}, \mathbb{I}/P)$, where ζ is a p^t th root of unity. Thus, by Deligne's theorem there is an irreducible Galois representation ρ_{ϕ} over $\text{Frac}(\mathbb{I}/P)$ unramified away from Np with

$$\begin{aligned} \det(1 - \rho_{\phi}(\text{Frob}_{\ell})T) &= 1 - \phi(a_{\ell})T + \psi\psi_{\zeta}\omega^{-k}(\ell)\ell^{k-1}T^2 \\ &= 1 - \phi(a_{\ell})T + \psi\psi_{\zeta}\omega^{-k}\chi^k(\ell)T^2 \\ &= 1 - \phi(a_{\ell})T + \psi(\ell)\phi(\Psi(\text{Frob}_{\ell}))T^2 \end{aligned}$$

Thus, ρ_{ϕ} is a residual representation as desired. $\square$

It turns out that the above will be sufficient to construct an $\mathbb{I}$ -adic representation attached to $\mathbf{f}$. The proof strategy will be to show that the existence of infinitely many residual representations implies the existence of the $\mathbb{I}$ -adic representation. Note that if $\mathbf{f}$ is such that the specializations $\phi_{1,\zeta}(f)$ are all cusp forms, then the full power of Deligne's theorem is not needed; instead, Eichler-Shimura's more classical method of finding Galois representations attached to weight 2 eigenforms in Jacobians of modular curves would be enough (see [Di05, Thm 9.5.4]).

Proof (Theorem 9). We follow the strategy of [Wi88]. We will proceed using the notion of a pseudo-representation from deformation theory. Let $j \in G_{\mathbb{Q}}$ be the complex conjugation. For our purposes, a pseudo-representation T of $G_{\mathbb{Q}}$ into a topological ring R will mean the data of continuous functions

$$a, d : G_{\mathbb{Q}} \rightarrow R, \quad x : G_{\mathbb{Q}} \times G_{\mathbb{Q}} \rightarrow R$$

Satisfying the following:

1. $a(\sigma\tau) = a(\sigma)a(\tau) + x(\sigma, \tau)$ and $d(\sigma\tau) = d(\sigma)d(\tau) + x(\tau, \sigma)$
2. $x(\sigma_1\tau_1, \sigma_2\tau_2) = a(\sigma_1)a(\tau_2)x(\tau_1, \sigma_2) + a(\tau_2)d(\tau_1)x(\sigma_1, \sigma_2) + a(\sigma_1)d(\sigma_2)x(\tau_1, \tau_2) + d(\tau_1)d(\sigma_2)x(\sigma_1, \tau_2)$
3. $a(1) = d(1) = d(j) = 1$, $a(j) = -1$, and $x(\sigma, \rho) = x(\rho, \tau) = 0$ for $\rho \in \{1, j\}$.
4. $x(\sigma_1, \tau_1)x(\sigma_2, \tau_2) = x(\sigma_1, \tau_2)x(\sigma_2, \tau_1)$

The origin of these formulae is not so arcane: these are the relations satisfied by a two dimensional representation

$$\sigma \mapsto \begin{pmatrix} a_{\sigma} & b_{\sigma} \\ c_{\sigma} & d_{\sigma} \end{pmatrix}$$

where $a(\sigma) = a_{\sigma}$, $d(\sigma) = d_{\sigma}$, and $x(\sigma, \tau) = b(\sigma)c(\tau)$, and where the given basis for our representation consists of a -1 eigenvector and a $+1$ eigenvector for j (in that order).

For each prime P of $\mathbb{I}$ as in Proposition 6, we have the corresponding residual representation ρ_P . Since $\det(\rho_P(\sigma))$ is given by an odd character on all but finitely many Frobenii, it follows by Chebotarev density that $\det(\rho_P(j)) = -1$. Thus, ρ_P has a basis given by a -1 and $+1$ eigenvector for $\rho_P(j)$, and so we may attach a pseudo-representation to each ρ_P . Note that the resulting matrix coefficients for ρ_P a priori may not be integral; but since $G_{\mathbb{Q}}$ is compact, ρ_P can be conjugated to have image in $\mathrm{GL}_2(\mathcal{O}_L)$, where L is the p -adic field of definition of ρ_P (i.e., the field of definition of the corresponding classical eigenform).

Moreover, note that the pseudo-representation so obtained depends only on the trace of ρ_P , since we may define our pseudo-representation by

$$a(\sigma) = \frac{1}{2}(\mathrm{Tr}(\sigma) + \mathrm{Tr}(\sigma j)), \quad d(\sigma) = \frac{1}{2}(\mathrm{Tr}(\sigma) - \mathrm{Tr}(\sigma j)), \quad x(\sigma, \tau) = a(\sigma\tau) - a(\sigma)a(\tau)$$

so since all but finitely many Frobenius traces for ρ_P lie in $\phi(\mathbb{I})$ where ϕ is as in Proposition 6, it follows that our pseudo-representation is valued in $\mathbb{I}/P \subset \mathcal{O}_L$.

For a pseudo-representation $T = (a, d, x)$, we define $\mathrm{Tr}(\sigma) = a(\sigma) + d(\sigma)$ and $\det(\sigma) = a(\sigma)d(\sigma) - x(\sigma, \sigma)$.

Our goal will be to patch together each of the $\mathbb{I}/P$ -valued pseudo-representations into a global $\mathbb{I}$ -valued pseudo-representation. Then, we will show that from this we can construct a genuine representation of $G_{\mathbb{Q}}$ with all the right properties.

Enumerate the primes P of $\mathbb{I}$ as in Proposition 6 by $P_1, P_2, \dots$. Suppose that we have a pseudo-representation $T_n = (a, d, x)$ into $\mathbb{I}/\bigcap_{i=1}^n P_i$ so that

$$\mathrm{Tr}(T_n(\mathrm{Frob}_{\ell})) \equiv a_{\ell} \pmod{\bigcap_{i=1}^n P_i}$$

for all $\ell \nmid Np$. Abusing notation, let $T_{P_{n+1}}$ denote the $\mathbb{I}/P_{n+1}$ -valued pseudo-representation constructed above. Considering the exact sequence

$$0 \rightarrow \mathbb{I} / \bigcap_{i=1}^{n+1} P_i \rightarrow \left(\mathbb{I} / \bigcap_{i=1}^n P_i \right) \oplus \mathbb{I} / P_{n+1} \xrightarrow{(a,b) \mapsto a-b} \mathbb{I} / \left(\bigcap_{i=1}^n P_i, P_{n+1} \right) \rightarrow 0$$

we see that the $(\mathbb{I} / \bigcap_{i=1}^n P_i) \oplus \mathbb{I} / P_{n+1}$ -valued pseudo-representation $T_n \oplus T_{P_{n+1}}$ vanishes on almost every Frobenius on the right, and thus by continuity is identically 0. We thereby obtain a pseudo-representation valued in $\mathbb{I} / \bigcap_{i=1}^{n+1} P_i$. Taking the limit, we obtain a pseudo-representation $\tilde{T}$ valued in

$$\varprojlim_n \mathbb{I} / \bigcap_{i=1}^n P_i = \mathbb{I}$$

satisfying

$$\mathrm{Tr}(\tilde{T}(\mathrm{Frob}_\ell)) = a_\ell, \quad \det(\tilde{T}(\mathrm{Frob}_\ell)) = \psi\psi(\mathrm{Frob}_\ell)$$

Now we want to see that our pseudo-representation can be upgraded to a genuine representation.

Let $\tilde{T} = (a, d, x)$. If $x(\sigma, \tau) = 0$ for all σ, τ , then

$$\sigma \mapsto \begin{pmatrix} a(\sigma) & \\ & d(\sigma) \end{pmatrix}$$

is such a representation and we are done. Otherwise, fix $\rho, \gamma \in G_{\mathbb{Q}}$ so that $x(\rho, \gamma) \neq 0$. Then, putting $b(\sigma) = x(\sigma, \gamma) / x(\rho, \gamma)$ and $c(\sigma) = x(\rho, \sigma)$, we may define $\rho : G_{\mathbb{Q}} \rightarrow \mathrm{Mat}_{2 \times 2}(\mathbb{K})$ by

$$\sigma \mapsto \begin{pmatrix} a(\sigma) & b(\sigma) \\ c(\sigma) & d(\sigma) \end{pmatrix}$$

This map has $\rho(1) = 1$, and it can be easily checked using the defining properties of pseudo-representations that ρ is then a homomorphism $G_{\mathbb{Q}} \rightarrow \mathrm{GL}_2(\mathbb{K})$. ρ clearly has the right trace and determinant.

The irreducibility and continuity of ρ remain to be checked. Let v_1, v_2 be the standard basis for $\mathbb{K}^2$ (i.e. the one in which ρ is given), and let $\mathcal{B} \subset \mathbb{K}^2$ be the $\mathbb{I}$ module generated by all $b(\sigma)v_1$ as σ ranges in $G_{\mathbb{Q}}$. Giving $\mathcal{B}$ the $\mathfrak{m}$ -adic topology, we see that $\mathcal{B}$ must be finite over $\mathbb{I}$ since $\mathbb{I}[G_{\mathbb{Q}}]$ is compact and $\sigma \mapsto b(\sigma)x(\rho, \gamma)$ is continuous. Then $M = \mathbb{I}v_2 \oplus \mathcal{B}v_1$ is a finite $\mathbb{I}$ -module stable under $G_{\mathbb{Q}}$ and which spans π over $\mathrm{Frac}(\mathbb{I})$, and the induced map $G_{\mathbb{Q}} \rightarrow \mathrm{End}_{\mathbb{I}}(M)$ is continuous since a, d , and x are continuous.

To show that ρ is irreducible, merely note that for any prime P used above in the construction of ρ , (the semisimplification of) M/PM is isomorphic to ρ_P , which is irreducible by Theorem 10, so M/PM must already be semisimple and irreducible. It follows that ρ must be irreducible, as desired. This completes the proof of Theorem 9. $\square$

In the proof of Theorem 9 it is quite visible how the representation ρ attached to $\mathbf{f}$ is given by interpolating together the representations attached to each specialization of $\mathbf{f}$, and how the representations attached to the specializations are recovered from ρ using the $\mathbb{I}$ -structure. This completes the philosophical picture underlying our dream of using the proof method of Ribet's converse to attack the Main Conjecture: we have now realized a “ p -adic family of Galois representations” over $\mathbb{I}$.

For our proof of the Main Conjecture, we will require one more auxiliary fact about the representation ρ , which we will state without proof.

Theorem 11. *Let ρ be the Galois representation attached to a normalized eigenform $\mathbf{f} \in \mathcal{S}^{\mathrm{ord}}(\psi, \mathbb{I})$. Let N be the conductor of ψ , and ℓ a prime with $\ell \mid Np$. Then if $a_\ell(\mathbf{f}) \neq 0$, we have*

$$\rho|_{D_\ell} \simeq \begin{pmatrix} \alpha_\ell \psi \Psi & * \\ & \alpha_\ell^{-1} \end{pmatrix}$$

where D_ℓ denotes a decomposition subgroup for ℓ and α_ℓ is unramified at ℓ with $\alpha_\ell(\mathrm{Frob}_\ell) = a_\ell(\mathbf{f})$.

For a proof, see [Wi86]. Let us justify a slight strengthening that we will need later:

Lemma 5. *In the setting of Theorem 11, suppose further that $\ell \neq p$. Then we in fact have*

$$\rho|_{D_\ell} \simeq \begin{pmatrix} \alpha\psi\Psi & \\ & \alpha^{-1} \end{pmatrix}$$

In particular, ρ has finite image on I_ℓ .

Proof. First, we record that $\alpha_\ell^2\Psi(\text{Frob}_\ell)$ is not a constant function in $\mathbb{I}$, in the sense that if we consider Λ_ψ as $\mathcal{O}_\psi[[T]]$, then this element of $\mathbb{I}$ does not lie in $\Lambda_\psi \setminus (T)$. Indeed, if this were the case, say with $\alpha_\ell^2\Psi(\text{Frob}_\ell)$ equal to a constant c , then for each specialization $f_{k,\zeta} = \phi_{k,\zeta}(\mathbf{f})$ we would have

$$a_\ell(f_{k,\zeta}) = \phi_{k,\zeta}(\alpha_\ell(\text{Frob}_\ell)) = \phi_{k,\zeta}(c\Psi(\text{Frob}_\ell)^{-1})^{1/2} = c^{1/2}\ell^{-k/2}$$

But $f_{k,\zeta}$ is a normalized eigenform and so should have integral Fourier coefficients, which is violated by the above for $k \gg 0$.

Now, note that $\alpha_\ell\psi\Psi(\text{Frob}_\ell) \neq \alpha_\ell^{-1}(\text{Frob}_\ell)$, since otherwise we would have $\alpha^2\Psi(\text{Frob}_\ell) = \psi(\text{Frob}_\ell)^{-1}$ which is a constant in Λ_ψ . Thus, $\rho(\text{Frob}_\ell)$ is diagonalizable and without loss of generality we may assume that $\ast(\text{Frob}_\ell) = 0$.

Let P_ℓ be the wild inertia subgroup of I_ℓ , and let K be the kernel of ρ . Since P_ℓ is pro- ℓ and $\text{GL}_2(\mathbb{K})$ is pro- p , P_ℓ/K must be finite and of ℓ -power order. Choose m sufficiently large so that $\ell^m \geq \#(P_\ell/K)$. Then for $\sigma \in I_\ell$ we have

$$\rho(\text{Frob}_\ell \sigma^{\ell^m} \text{Frob}_\ell^{-1}) = \begin{pmatrix} \psi(\sigma^{\ell^m}) \alpha^2\psi\Psi(\text{Frob}) \cdot \ast(\sigma^{\ell^m}) & \\ & 1 \end{pmatrix}$$

On the other hand, the conjugation action of Frob on I_ℓ induces the ℓ -power map on I_ℓ/P_ℓ . Thus, since π factors through tame inertia on ℓ^m th powers, we also have

$$\rho(\text{Frob}_\ell \sigma^{\ell^m} \text{Frob}_\ell^{-1}) = \rho(\sigma^{\ell^{m+1}}) = \begin{pmatrix} \psi(\sigma^{\ell^{m+1}}) \ast(\sigma^{\ell^{m+1}}) & \\ & 1 \end{pmatrix} = \begin{pmatrix} \psi(\sigma^{\ell^{m+1}}) (1 + \psi(\sigma^{\ell^m}) + \dots + \psi^{\ell-1}(\sigma^{\ell^m})) \cdot \ast(\sigma^{\ell^m}) & \\ & 1 \end{pmatrix}$$

Thus, we have

$$\left(\alpha^2\psi\Psi(\text{Frob}) + \sum_{i=1}^{\ell-1} \psi(\sigma^{\ell^m})^i \right) \cdot \ast(\sigma^{\ell^m}) = 0$$

Since $\alpha^2\psi\Psi(\text{Frob})$ is not a constant, it follows that $\ast(\sigma^{\ell^m}) = 0$.

To conclude, observe that since P_ℓ/K is finite and since $I_\ell/P_\ell \simeq \prod_{q \neq \ell} \mathbb{Z}_q$, the subgroup H of I_ℓ generated by ℓ^m th powers is finite index in I_ℓ/K . Then the map

$$H^1(I_\ell/K, \mathbb{K}^2) \xrightarrow{\text{res}} H^1(H/K, \mathbb{K}^2) \xrightarrow{\text{cores}} H^1(I_\ell/K, \mathbb{K}^2)$$

is given by multiplication by $[I_\ell/K : H/K]$, which is invertible since we are working over the characteristic 0 field $\mathbb{K}$. Since $\ast$ is trivial in $H^1(H/K, \mathbb{K}^2)$, it follows that $\ast$ is trivial in $H^1(I_\ell/K, \mathbb{K}^2)$ as well, and the result follows. $\square$

6 Proof of the Main Conjecture

This section will be devoted entirely to the proof of Theorem 7. We follow the method of [Sk09]. We will aim to prove that

$$\text{ord}_P(f_\psi) = \text{ord}_P(g_{\psi^{-1}})$$

for each height 1 prime P of Λ_ψ . Since each such prime is principal, we have

$$\lambda(r) = \sum_P \text{ord}_P(r) \lambda(P), \quad \mu(r) = \sum_P \text{ord}_P(r) \mu(P).$$

Thus, in light of Theorem 6, it suffices to prove that

$$\text{ord}_P(f_\psi) \geq \text{ord}_P(g_{\psi^{-1}})$$

for each P .

For this section, let N_ψ be the conductor of ψ , and write $N_\psi = Np^r$ for N prime to p . Our proof will utilize the Λ_ψ -adic Eisenstein family $\mathcal{E}_\psi$ defined in Section 5. Let $\mathbb{T}_\psi$ be the ordinary Hecke algebra for Λ_ψ -adic cusp forms of nebentypus ψ . Then we can define an ideal I_ψ of $\mathbb{T}_\psi$ by

$$I_\psi := \left(\{T_\ell - 1 - \psi\Psi(\text{Frob}_\ell) : \ell \nmid Np\}, \{U_\ell - 1 : \ell \mid Np\} \right).$$

This is called the Eisenstein ideal of $\mathbb{T}_\psi$, because it looks like the relations determined by how the Hecke operators should act on $\mathcal{E}_\psi$ (even though $\mathcal{E}_\psi$ is not a cusp form). As in the proof of Ribet's converse, our strategy will be to use the Eisenstein ideal to cut out a cusp form which looks like $\mathcal{E}_\psi \pmod{P}$.

The composite $\Lambda_\psi \rightarrow \mathbb{T}_\psi \rightarrow \mathbb{T}_\psi/I_\psi$ is clearly surjective; denote its kernel by J_ψ .

Theorem 12. *Let P be a height one prime of Λ_ψ so that $p \notin P$ and $\psi\Psi|_{G_{\mathbb{Q}_p}} \not\equiv 1 \pmod{P}$. Then*

$$\text{ord}_P(J_\psi) \geq \text{ord}_P(g_\psi).$$

Proof. Note that if ψ is of the form $\omega^{-1}\psi_\zeta$, then g_ψ is a unit by Lemma 2 and there is nothing to prove; more generally, if $P \nmid g_\psi$, then there is nothing to prove, so assume $P \mid g_\psi$ (and so in particular ψ is not of the above form).

Choose some even $k_0 > 2$ divisible by $p-1$ so that $P \nmid g_\psi(\gamma\chi^{-k_0}(\gamma)-1)$ (which is possible, say, by Weierstrass preparation, so the polynomial part of P has finitely many roots). We write

$$\mathcal{G}_\psi := e(G_{k_0}\mathcal{E}_\psi(\gamma\chi^{-k_0}(\gamma)-1)) = g_\psi(\gamma\chi^{-k_0}(\gamma)-1) + \sum_{n=1}^{\infty} b(n)q^n$$

where e is the ordinary projector and G_{k_0} is the modular form defined in section 2. Note that $\mathcal{G}_\psi$ is a Λ_ψ -adic modular form and that the right hand side has the correct constant term since G_{k_0} is weight k_0 with constant term 1. We may then define

$$\mathcal{F} := \prod_{\ell \mid N} (U_\ell - \Psi(\text{Frob}_\ell)) \left(g_\psi(\gamma\chi^{-k_0}(\gamma)-1)\mathcal{E}_\psi - g_\psi\mathcal{G}_\psi \right) = \sum_{n=1}^{\infty} c(n)q^n.$$

It can be checked that applying $\prod (U_\ell - \Psi(\text{Frob}_\ell))$ to an ordinary form with no constant term at the ∞ -cusp produces an ordinary cusp form (see [Wi90, Lemma 3.2]), so $\mathcal{F}$ is a cusp form.

Because U_ℓ as an operator on the full space $\mathcal{M}^{\text{ord}}(\psi, \Lambda_\psi)$ fixes $\mathcal{E}_\psi$, we have for primes q

$$c(q) \equiv \prod_{\ell \mid N} (1 - \Psi(\text{Frob}_\ell)) g_\psi(\gamma\chi^{-k_0}(\gamma)-1) \begin{cases} (1 + \psi\Psi(\text{Frob}_q)) & q \nmid Np \\ 1 & q \mid Np \end{cases} \pmod{P^{\text{ord}_P(g_\psi)}}.$$

Also note that we have

$$c(1) \equiv \prod_{\ell \mid N} (1 - \Psi(\text{Frob}_\ell)) g_\psi(\gamma\chi^{-k_0}(\gamma)-1) \pmod{P^{\text{ord}_P(g_\psi)}}.$$

By assumption we have $g_\psi(\gamma\chi^{-k_0}(\gamma)-1) \notin P$. Moreover, suppose for contradiction that we had $1 - \Psi(\text{Frob}_\ell) \in P$. Then $\Psi(\gamma)$ is a root of unity mod P , since $\Psi(\text{Frob}_\ell)$ is a p -adic power of γ and any quotient of Γ by a closed subgroup is finite. So taking $\Psi(\gamma) \equiv \xi \pmod{P}$ for a root of unity ξ , the map $\Lambda_\psi \rightarrow \Lambda_\psi/P$ factors through the map $\Lambda_\psi \rightarrow \overline{\mathbb{Q}}_p$ sending γ to ξ , which takes g_ψ to $L^{\{p\}}(0, \psi\psi_\xi)$. Thus, since $P \mid g_\psi$, $L^{\{p\}}(0, \psi\psi_\xi)$

must map to 0 in Λ_ψ/P . But $p \notin P$ by assumption, so we must have $L^{\{p\}}(0, \psi\psi_\xi) = 0$. Since $\psi\psi_\xi$ is odd, $L(0, \psi\psi_\xi)$ is nonzero, so the p -factor must vanish at 0, giving

$$1 - \psi\psi_\xi(p)p^0 = 0 \implies \psi(p) = \psi_\xi^{-1}(p).$$

This implies that $\psi\psi_\xi$ is unramified at p (lest the p -factor be identically 1), so ψ and ψ_ξ^{-1} agree on I_p ; and from the above we see that ψ and ψ_ξ^{-1} agree on a Frobenius for p . Thus, we have

$$\psi|_{G_{\mathbb{Q}_p}} = \psi_\xi^{-1}|_{G_{\mathbb{Q}_p}} \equiv \Psi^{-1}|_{G_{\mathbb{Q}_p}} \pmod{P},$$

contradicting our assumptions. Thus, we must have $1 - \Psi(\text{Frob}_\ell) \notin P$.

Thus, dividing out by the mod P unit $c(1)$, we have that T_ℓ acts on $\mathcal{F}$ by $1 + \psi\Psi(\text{Frob}_\ell) \pmod{P}$ (since this is how T_ℓ acts on $\mathcal{E}_\psi$), and U_ℓ acts by 1. It follows that the map $\mathbb{T}_{\psi,P} \rightarrow \Lambda_{\psi,P}/P^{\text{ord}_P(g_\psi)}$ given by

$$T_\ell \mapsto 1 + \psi\Psi(\text{Frob}_\ell), \quad U_\ell \mapsto 1$$

is a well-defined $\Lambda_{\psi,P}$ -algebra homomorphism. Obviously the kernel of this map contains $I_\psi\mathbb{T}_{\psi,P}$, and thus also contains J_ψ ; since this map is clearly surjective, the theorem follows. $\square$

To compare J_ψ to (f_ψ) , we will utilize the Λ_ψ -adic Galois representations coming from the eigenforms attached to I_ψ . As in Ribet's converse, we will use these representations to construct a non-split mod P extension of Galois modules, which gives an element in cohomology. We will then use this class to get a handle on the cohomology group $\text{Sel}_\infty(\psi)$.

Theorem 13. *Let P be a height one prime of Λ_ψ so that $p \notin P$ and $\psi\Psi|_{G_{\mathbb{Q}_p}} \not\equiv 1 \pmod{P}$. Then*

$$\text{ord}_P(\text{Ch}(X_\infty(\psi^{-1}))) \geq \text{ord}_P(J_\psi).$$

Proof. As in the previous theorem, assume without loss of generality that $\text{ord}_P(J_\psi) \geq 1$. Since $\mathbb{T}_\psi$ is finite over Λ_ψ and $\Lambda_\psi \cap I_\psi = 0$, by going up there is a unique height one prime P' of $\mathbb{T}_\psi$ containing (I_ψ, P) . Let $Q_1, \dots, Q_m$ be the minimal primes of $\mathbb{T}_\psi$ contained in P' ; let $\mathbb{J}_i := \mathbb{T}_\psi/Q_i$ be the i th component of $\mathbb{T}_\psi$, and let $A := \mathbb{T}_\psi / \cap_i Q_i$. Note that we have $A_{P'} = \mathbb{T}_{\psi,P'}$; this follows from the fact that $\mathbb{T} \otimes_{\Lambda_\psi} \text{Frac}(\Lambda_\psi)$ is semisimple. It follows that

$$A_P/I_\psi A_P = A_{P'}/I_\psi A_{P'} = \mathbb{T}_{\psi,P'}/I_\psi \mathbb{T}_{\psi,P'} = \Lambda_{\psi,P}/J_\psi \Lambda_{\psi,P}.$$

Write $K := A \otimes_{\Lambda_\psi} \text{Frac} \Lambda_\psi = \prod_i \text{Frac}(\mathbb{J}_i)$. Let $\rho_i : G_{\mathbb{Q}} \rightarrow \text{GL}(V_i)$ be the $\mathbb{J}_i$ -adic Galois representation attached to (the eigenform $\mathbf{f}_i$ corresponding to) Q_i . Note that the ρ_i are pairwise non-isomorphic: if ρ_i were isomorphic to ρ_j , then we would have $a_\ell(\mathbf{f}_i) = a_\ell(\mathbf{f}_j)$ for all $\ell \nmid Np$. It is known that the $\mathbf{f}_i$ specialize to forms new away from p ([Hi00][Cor. 3.7]), so it would follow from strong multiplicity one for newforms that $a_\ell(\mathbf{f}_i) = a_\ell(\mathbf{f}_j)$ for $\ell \mid Np$, whence $\mathbf{f}_i = \mathbf{f}_j$.

We can write $\rho = \bigoplus_i \rho_i$, which is a K -linear Galois representation on the free rank-2 K -module $V = \bigoplus_i V_i$. Then for $\ell \nmid Np$, $\text{Tr}(\rho(\text{Frob}_\ell))$ is the image of T_ℓ in A because $\text{Tr}(\rho_i(\text{Frob}_\ell))$ is the image of T_ℓ in each $\mathbb{J}_i$. Thus, by Chebotarev density and continuity of our representation, we have that the trace of ρ is valued in $A \subset K$ over all of $G_{\mathbb{Q}}$. In fact, we have

$$\text{Tr} \rho \equiv \psi\Psi + 1 \pmod{I_\psi A}$$

since this is true of the image of each T_ℓ by the definition of I_ψ .

By assumption, there is $\sigma_0 \in G_{\mathbb{Q}_p}$ with $\psi\Psi(\sigma_0) - 1 \notin P$. Since the $\mathbf{f}_i$ are ordinary, ρ can be made upper triangular on $G_{\mathbb{Q}_p}$ by Theorem 11; choose a K -basis $\{v_1, v_2\}$ of V so that

$$\rho|_{G_{\mathbb{Q}_p}} = \begin{pmatrix} \alpha^{-1} \psi\Psi & * \\ & \alpha \end{pmatrix}$$

with α unramified at p , $\alpha(\text{Frob}_p)$ the image of U_p in A , and so that $\rho(\sigma_0)$ is diagonal. In this basis, let us write

$$\rho(\sigma) = \begin{pmatrix} a_\sigma & b_\sigma \\ c_\sigma & d_\sigma \end{pmatrix}, \quad \sigma \in G_{\mathbb{Q}}.$$

Put $r := \alpha^{-1}\psi\Psi(\sigma_0) - \alpha(\sigma_0)$. Observe that since $\alpha(\text{Frob}_p) = U_p \equiv 1 \pmod{I_\psi A}$, α is the trivial character mod I_ψ , so we have

$$r \equiv \psi\Psi(\sigma_0) - 1 \not\equiv 0 \pmod{(I_\psi, P)}.$$

Now let us compute the matrix coefficients of $\rho(\sigma) \bmod I_\psi$. Let $\delta_1 := \sigma_0 - \alpha(\sigma_0)$ and $\delta_2 := \sigma_0 - \alpha^{-1}\psi\Psi(\sigma_0)$ be elements of $A[G_{\mathbb{Q}}]$. Then we have

$$\begin{aligned} \text{Tr}(\rho(\delta_1\sigma)) &= \text{Tr}(\rho(\sigma_0\sigma) - \alpha(\sigma_0)\rho(\sigma)) \\ &= \text{Tr} \left(\begin{pmatrix} a_\sigma \alpha^{-1}\psi\Psi(\sigma_0) & b_\sigma \alpha^{-1}\psi\Psi(\sigma_0) \\ c_\sigma \alpha(\sigma_0) & d_\sigma \alpha(\sigma_0) \end{pmatrix} - \begin{pmatrix} \alpha(\sigma_0)a_\sigma & \alpha(\sigma_0)b_\sigma \\ \alpha(\sigma_0)c_\sigma & \alpha(\sigma_0)d_\sigma \end{pmatrix} \right) \\ &= a_\sigma(\alpha^{-1}\psi\Psi(\sigma_0) - \alpha(\sigma_0)) = r a_\sigma. \end{aligned}$$

Then we have

$$\begin{aligned} \text{Tr}(\delta_1\sigma) &= \psi\Psi(\sigma_0\sigma) + 1 - \alpha(\sigma_0)\psi\Psi(\sigma) - \alpha(\sigma_0) \\ &\equiv (\psi\Psi(\sigma_0) - 1)\psi\Psi(\sigma) \pmod{I_\psi} \\ &= r\psi\Psi(\sigma). \end{aligned}$$

Similarly, for d_σ we have

$$\begin{aligned} \text{Tr}(\rho(\delta_2\sigma)) &= \text{Tr}(\rho(\sigma_0\sigma) - \alpha^{-1}\psi\Psi(\sigma_0)\rho(\sigma)) \\ &= \text{Tr} \left(\begin{pmatrix} a_\sigma \alpha^{-1}\psi\Psi(\sigma_0) & b_\sigma \alpha^{-1}\psi\Psi(\sigma_0) \\ c_\sigma \alpha(\sigma_0) & d_\sigma \alpha(\sigma_0) \end{pmatrix} - \begin{pmatrix} \alpha^{-1}\psi\Psi(\sigma_0)a_\sigma & \alpha^{-1}\psi\Psi(\sigma_0)b_\sigma \\ \alpha^{-1}\psi\Psi(\sigma_0)c_\sigma & \alpha^{-1}\psi\Psi(\sigma_0)d_\sigma \end{pmatrix} \right) \\ &= d_\sigma(\alpha(\sigma_0) - \alpha^{-1}\psi\Psi(\sigma_0)) = -r d_\sigma, \end{aligned}$$

and then

$$\begin{aligned} \text{Tr}(\rho(\delta_2\sigma)) &\equiv \psi\Psi(\sigma_0\sigma) + 1 - \alpha^{-1}\psi\Psi(\sigma_0)\psi\Psi(\sigma) - \alpha^{-1}\psi\Psi(\sigma_0) \pmod{I_\psi} \\ &\equiv 1 - \psi\Psi(\sigma_0) \pmod{I_\psi} \\ &\equiv -r \pmod{I_\psi}. \end{aligned}$$

Finally, note that the relation $a_{\sigma\tau} = a_\sigma a_\tau + b_\sigma c_\sigma$ implies

$$r^2 b_\sigma c_\sigma \equiv 0 \pmod{I_\psi}.$$

Since $r \notin (I_\psi, P)$, it follows from all of the above that

$$a_\sigma \equiv \psi\Psi(\sigma), \quad d_\sigma \equiv 1, \quad b_\sigma d_\sigma \equiv 0 \pmod{I_\psi A_P}.$$

We will use these congruences to exhibit a certain non-split Galois stable lattice in V . We make note of the fact that the A -module $\mathcal{C}$ consisting of all c_σ for $\sigma \in A[G_{\mathbb{Q}}]$ is finite since ρ is continuous and $G_{\mathbb{Q}}$ is compact.

Let $\mathcal{M} := A_P v_1 \oplus \mathcal{C}_P v_2$, an A_P -lattice in V (though not necessarily free). $\mathcal{M}$ is $G_{\mathbb{Q}}$ -stable, and in fact is generated as an $A_P[G_{\mathbb{Q}}]$ -module by v_1 . $\mathcal{M} \otimes_{A_{\psi, P}} \text{Frac}(\Lambda_\psi)$ is a nontrivial $K[G_{\mathbb{Q}}]$ -stable submodule of V which projects nontrivially onto each V_i (say, since $v_1 \in \mathcal{M}$). Thus, by the irreducibility and distinctness of the V_i , we must have $\mathcal{M} \otimes_{A_{\psi, P}} \text{Frac}(\Lambda_\psi) = V$. In particular, since $A \hookrightarrow K$ (A is reduced), we see that $\mathcal{C}$ is a faithful A -module.

It is from $\mathcal{M}$ that we will construct an extension giving a suitable element of cohomology. Let $\mathcal{M}_1 = A_P v_1$ and $\mathcal{M}_2 = \mathcal{C}_P v_2$ so that $\mathcal{M} = \mathcal{M}_1 \oplus \mathcal{M}_2$. For $cv_2 \in \mathcal{M}_2$, using the above mod I_ψ congruences in A_P we see that

$$\rho(\sigma)cv_2 = b_\sigma cv_1 \oplus d_\sigma cv_2 \in cv_2 + I_\psi \mathcal{M}.$$

Thus, letting $\overline{\mathcal{M}}$ and $\overline{\mathcal{M}}_2$ be the reductions of $\mathcal{M}$ and $\mathcal{M}_2$ modulo I_ψ , we find that $\overline{\mathcal{M}}_2$ is an $A_P[G_\mathbb{Q}]$ -submodule of $\overline{\mathcal{M}}$ with trivial $G_\mathbb{Q}$ -action. We also then see via the above congruences that $\overline{\mathcal{M}}_1 := \overline{\mathcal{M}}/\overline{\mathcal{M}}_2$ is a rank 1 $A[G_\mathbb{Q}]$ -module acted upon by $G_\mathbb{Q}$ via the character $\psi\Psi$. Moreover, we clearly see that $\mathcal{M}_1$ surjects onto $\overline{\mathcal{M}}_1$.

It follows that the extension

$$0 \rightarrow \overline{\mathcal{M}}_2 \rightarrow \overline{\mathcal{M}} \rightarrow \overline{\mathcal{M}}_1 \rightarrow 0$$

is a non-split extension of $A_P/I_\psi A_P[G_\mathbb{Q}] = \Lambda_{\psi,P}/J_\psi \Lambda_{\psi,P}[G_\mathbb{Q}]$ modules, with non-splitness following from the fact that v_1 generates $\mathcal{M}$ as an $A_P[G_\mathbb{Q}]$ -module. However, this extension is split over inertia for every prime: by our choice of basis, $c_\sigma = 0$ for $\sigma \in I_p$. For primes $\ell \mid N$ away from p , $U_\ell - 1 \in P$, so the image of U_ℓ in $\mathbb{J}_i$ is nonzero. Thus the U_ℓ -eigenvalue for $\mathbf{f}_i$ is nonzero, so we may use the fact that $\rho_i|_{G_{\mathbb{Q}_\ell}}$ can be written as the direct sum of characters $\beta^{-1}\psi\Psi$ and β where β is unramified. Then since Ψ is unramified at ℓ , it follows that $\rho|_{I_\ell}$ factors through ψ . In particular, $\ker \rho \cap I_\ell$ is finite index in I_ℓ , and so

$$H^1(I_\ell, \overline{\mathcal{M}}) \xrightarrow{res} H^1(\ker \rho \cap I_\ell, \overline{\mathcal{M}}) \xrightarrow{cores} H^1(I_\ell, \overline{\mathcal{M}})$$

is given by multiplication by $[I_\ell : I_\ell \cap \ker \rho]$. Since $p \notin P$, all primes are invertible in A_P/I_ψ , so this map is invertible. Then since $*|_{I_\ell \cap \ker \rho}$ is split, it follows that $*|_{I_\ell}$ is split.

Letting $\bar{v}_1$ be the image of v_1 in $\overline{\mathcal{M}}$, we can now define $\mathfrak{M}$ to be the $A/I_\psi[G_\mathbb{Q}]$ -module generated by $\bar{v}_1$. Let $\mathfrak{M}_2 = \mathfrak{M} \cap \overline{\mathcal{M}}_2$, which is an $A/I_\psi[G_\mathbb{Q}]$ -submodule of $\mathfrak{M}$, and put $\mathfrak{M}_1 = \mathfrak{M}/\mathfrak{M}_2$. Then $\mathfrak{M}_1$ is the $A/I_\psi[G_\mathbb{Q}]$ -submodule of $\overline{\mathcal{M}}_1$ generated by the image of $\bar{v}_1$. From this, we see that $G_\mathbb{Q}$ acts trivially on $\mathfrak{M}_2$ and acts by $\psi\Psi$ on $\mathfrak{M}_1$.

Summing up the above, we have

$$\overline{\mathcal{M}}_1 \simeq A_P/I_\psi A_P(\psi\Psi) \simeq \Lambda_{\psi,P}/J_\psi \Lambda_{\psi,P}(\psi\Psi) = \Lambda_{\psi,P}/P^{\text{ord}_P(J_\psi)}(\psi\Psi),$$

with the last equality following since $\Lambda_{\psi,P}$ is a DVR. So then $\mathfrak{M}_1 \simeq \Lambda_{\psi,P}/P^{\text{ord}_P(J_\psi)}(\psi\Psi)$, whence we have a $\Lambda_{\psi,P}/P^{\text{ord}_P(J_\psi)}[G_\mathbb{Q}]$ -extension

$$0 \rightarrow \mathfrak{M}_2 \rightarrow \mathfrak{M} \rightarrow \Lambda_{\psi,P}/P^{\text{ord}_P(J_\psi)} \rightarrow 0$$

which is non-split since its localization at P is non-split. As such, we have produced a nonzero class

$$[\mathfrak{M}] \in H^1(\mathbb{Q}, \mathfrak{M}_2(\psi^{-1}\Psi^{-1})).$$

Define a map

$$\theta : \mathfrak{M}_2^* = \text{hom}_{\Lambda_\psi}(\mathfrak{M}_2, \Lambda_\psi^*) \rightarrow H^1(\mathbb{Q}, \Lambda_\psi^*(\psi^{-1}\Psi^{-1}))$$

by sending a homomorphism φ to the class given by pushing forward $[\mathfrak{M}]$ along φ . Consider now the P -localization of the dual map

$$\theta^* : H^1(\mathbb{Q}, \Lambda_\psi^*(\psi^{-1}\Psi^{-1}))_P^* \rightarrow \mathfrak{M}_{2,P}.$$

Observe that since $\overline{\mathcal{M}} = \mathfrak{M}_P$ is split over inertia at all primes, for each ℓ we can write

$$\psi^{-1}\Psi^{-1}(\sigma)\sigma\bar{v}_1 - \bar{v}_1 = \psi^{-1}\Psi^{-1}(g)c\bar{v}_2 - c\bar{v}_2$$

for all $\sigma \in I_\ell$ and some $c \in \mathcal{C}_P$. Choosing $r \in \Lambda_\psi \setminus P$ so that $rc\bar{v}_2 \in \mathfrak{M}_2$, it follows that $r[\mathfrak{M}]$ is unramified at ℓ . $[\mathfrak{M}]$ is already unramified at all places away from Np (since ρ is), so it follows that we may find $r \notin P$ with $r[\mathfrak{M}]$ everywhere unramified. It follows that θ_P factors through $\text{Sel}_\infty(\psi^{-1})_P$, and so θ_P^* factors through the surjection onto $X_\infty(\psi^{-1})$.

Next we will show that θ_P^* surjects onto $\mathfrak{M}_{2,P}$. Let $\mathfrak{M}'_2$ be the image of θ^* . Suppose for a moment that we had some $s \in \Lambda_\psi \setminus P$ so that $s[\mathfrak{M}]$ mapped to zero in $H^1(\mathbb{Q}, \mathfrak{M}_2/\mathfrak{M}'_2(\psi^{-1}\Psi^{-1}))$. Then it would follow that

$$\begin{array}{ccccc} 0 \rightarrow (\mathfrak{M}_2/\mathfrak{M}'_2)_P \rightarrow (\mathfrak{M}/\mathfrak{M}'_2)_P \rightarrow \mathfrak{M}_{1,P} \rightarrow 0 \\ \parallel & & \parallel & & \parallel \\ \overline{\mathcal{M}}_2/\overline{\mathcal{M}}'_{2,P} & & \overline{\mathcal{M}}/\overline{\mathcal{M}}'_{2,P} & & \overline{\mathcal{M}}_1 \end{array}$$

is split. However, this is impossible if $(\mathfrak{M}_2/\mathfrak{M}'_2)_P \neq 0$ since $\bar{v}_1$ generates $\overline{\mathcal{M}}$. Thus, given such an s , θ_P^* is surjective; so let us produce such an s .

For this, note that

$$\mathfrak{M}'_2 = \bigcap_{\phi \in \ker \theta} \ker \phi.$$

For $S \subset \ker \theta$ finite, write $\mathfrak{M}_S = \bigcap_{\phi \in S} \ker \phi$, so that

$$H^1(\mathbb{Q}, \mathfrak{M}_2/\mathfrak{M}'_2(\psi^{-1}\Psi^{-1})) = \varprojlim H^1(\mathbb{Q}, \mathfrak{M}_2/\mathfrak{M}_S(\psi^{-1}\Psi^{-1})).$$

Thus, we need only find an $s \notin P$ annihilating the image of $[\mathfrak{M}]$ in each $H^1(\mathbb{Q}, \mathfrak{M}_2/\mathfrak{M}_S(\psi^{-1}\Psi^{-1}))$. But this image of $[\mathfrak{M}]$ is in the kernel of the map

$$H^1(\mathbb{Q}, \mathfrak{M}_2/\mathfrak{M}_S(\psi^{-1}\Psi^{-1})) \rightarrow \prod_{\phi \in S} H^1(\mathbb{Q}, \Lambda_\psi^*(\psi^{-1}\Psi^{-1}))$$

induced by the injective map $\mathfrak{M}_2/\mathfrak{M}_S \rightarrow \prod_{\phi \in S} \Lambda_\psi$ given by $m \mapsto (\phi(m))_{\phi \in S}$. Considering the related long exact sequence on cohomology, we see that $[\mathfrak{M}]$ is in the image of the $G_{\mathbb{Q}}$ -invariants of a quotient of $\prod_{\phi \in S} \Lambda_\psi^*(\psi^{-1}\Psi^{-1})$. But such an image is killed by $1 - \psi\Psi(\sigma_0)$, which is not in P by assumption. Thus we have our desired s , whence θ_P^* is indeed surjective.

Now we are ready to finish the proof of Theorem 13. For this, we will use the theory of Fitting ideals; for a brief introduction, see [Stacks, Tag 07Z6]. It follows from the above that we have a surjection $X_\infty(\psi^{-1})_P \rightarrow \mathfrak{M}_{2,P}$. This justifies the third inequality in the following:

$$\begin{aligned} \text{ord}_P(\text{Ch}(X_\infty(\psi^{-1}))) &= \text{ord}_P(\text{Ch}(X_\infty(\psi^{-1}))_P) \\ &= \text{ord}_P(\text{Fitt}_{\Lambda_{\psi,P}} \Lambda_{\psi,P} / \text{Ch}(X_\infty(\psi^{-1}))_P) \\ &\geq \text{ord}_P(\text{Fitt}_{\Lambda_{\psi,P}} \mathfrak{M}_{2,P}) \\ &= \text{ord}_P(\text{Fitt}_{\Lambda_{\psi,P}} \overline{\mathcal{M}}_2). \end{aligned}$$

Also observe that $\text{Fitt}_A \mathcal{M}_2 = 0$ since $\mathcal{M}_2 = \mathcal{C}v_2$ is a faithful A -module. Then since Fitting ideals are compatible with base change, we have

$$0 = \text{Fitt}_{A_P/I_\psi} \overline{\mathcal{M}}_2 = \text{Fitt}_{\Lambda_{\psi,P}/J_\psi} \overline{\mathcal{M}}_2.$$

Again by base change, $\text{Fitt}_{\Lambda_{\psi,P}/J_\psi} \overline{\mathcal{M}}_2$ is the image of $\text{Fitt}_{\Lambda_{\psi,P}} \overline{\mathcal{M}}_2$ modulo J_ψ ; it follows that

$$\text{ord}_P(\text{Fitt}_{\Lambda_{\psi,P}} \overline{\mathcal{M}}_2) \geq \text{ord}_P(J_\psi),$$

which suffices for the proof. $\square$

Theorems 12 and 13 together determine our desired inequality for most primes of Λ_ψ ; it remains to show the cases where $p \in P$ and where $\psi\Psi|_{G_{\mathbb{Q}_p}} \equiv 1 \pmod{P}$. In the former case, Ferraro and Washington showed that $\text{ord}_P(g_\psi) = 0$, and in the latter case Ferraro and Greenberg showed that $\text{ord}_P(f_\psi) = 1 = \text{ord}_P(g_{\psi^{-1}})$.

Thus, we have

$$\text{ord}_P(f_\psi) \geq \text{ord}_P(g_\psi)$$

for all height 1 primes P . As we have already noted, the Main Conjecture follows.

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